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UNIVERSITY OF GALWAY

**Young System Development in a Cometary Globule:
A Study of Infrared and Radio Observations of the
Circumstellar Disk Around AT Pyx**

*A thesis submitted in fulfilment of the requirements for the
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Dan McLachlan

19443664

Supervisor Dr. Christian Ginski

d.mclachlan1@universityofgalway.ie

danmclachlan77@gmail.com

PHYSICS DEPARTMENT, SCHOOL OF NATURAL SCIENCES, UNIVERSITY OF GALWAY

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Abstract

Aims: To understand the formation of planetary systems, it is necessary to observe and study systems at different evolutionary stages and in different environments. This project focuses on the AT Pyx system, a protoplanetary disk-hosting young star located in a cometary globule in the Gum Nebula. This is a unique system in that its formation environment differs greatly from the types of environments disks have generally been observed in to date. Using a host of different sources of data this project aims to infer what processes are at play in the disk and what the prognosis may be for planet formation.

Methods: Using data from the VLT's (Very Large Telescope) instruments XSHOOTER, ESPRESSO and - most prominently - SPHERE along with data from ALMA (Atacama Large Millimetre-submillimetre Array), a variety of measurements (geometric, photometric and otherwise) are made to characterise the observed disk features such as spiral arms and eccentricity. Mapping of the velocity components is also undertaken using the ALMA gas line data to characterise disk orientation and determine likelihood that the system is experiencing a late-stage infall event.

Results: The disk is measured to have a position angle of $\sim 22^\circ$ and an inclination of $\sim 42.5^\circ$. The disk is found to be eccentric with tentative $e \approx 0.626$ when deprojected according to literature values for disk height profile. Under the assumption that the formation of a planet is wholly responsible for the primary and secondary spiral arms, the mass of such a planet is found to range between 0.02 and 3 Jupiter masses. Measurement of the velocities associated with nearby globule cloud material returns appropriate velocities for a late-stage infall event. Far-ultraviolet field strength at AT Pyx's location is found to be low in comparison to other surveyed disks. It is also found that AT Pyx is possibly a binary system.

Conclusions: AT Pyx was the first disk within a cometary globule to be spatially resolved, and is now the first such disk to be investigated to this extent. The work of this project could potentially be a first step into the study of disks in the moderate-FUV environment of the Gum Nebula and its globules. Further work involving the high-resolution imaging of AT Pyx and other such disks is recommended and it is suggested that AT Pyx is an excellent candidate target for the forthcoming ELT. A sample of other disks for study in the same region both inside and outside cometary globules and molecular clouds is presented.

1 Introduction

Observational studies of planet-formation require multi-layered case studies on star systems at different evolutionary stages and in different environments. In order to investigate the formation of planets as it occurs, the circumstellar (or protoplanetary) disk phase of system development is of particular interest.

This project is a case study on the T Tauri star AT Pyx, located within the cometary globule CG22 in the Gum nebula. A young ($5.1^{+1.5}_{-1.0}$ Myr, Ginski et al. 2022) $\sim 1.25M_{\odot}$ K-type star (Herczeg and Hillenbrand, 2014), it hosts a circumstellar, potentially planet-forming disk with an extent in scattered light of approximately 126 AU. With a range of data probing multiple layers of this disk, this project seeks to clarify what can be determined at this stage regarding AT Pyx’s disk’s capacity to form planets and other processes occurring within the disk. A major focus of the project is how its unique location affects its characteristics. To this end, this thesis presents the canonical context of star- and planet-formation theory and observation as it relates to AT Pyx, followed by all methods used to probe the disk’s features, concluded with the fruits of this endeavour and suggestions for future work.

1.1 Star Formation and Evolution

1.1.1 Molecular Clouds

Star formation occurs in cold, dense molecular clouds of dust and gas, with the term ‘cloud’ representing a partition of dust and gas separate from surrounding interstellar medium (ISM), although the categorization of how molecular clouds are separated from diffuse ISM is complicated by the existence of broad definitions for the ISM that include all regions of molecular clouds save for the densest cores in which star-formation occurs (Hennebelle and Falgarone, 2012).

Their boundaries are commonly defined by regions of CO or dust emission (Kennicutt and Evans, 2012) and the matter within can largely be considered neutral due to the self-shielding mechanisms (generally from the absorption of photons by dust grains and CO molecules) at these boundaries which protect the inner material from ionisation (Krumholz, 2014). While they consist mostly of H_2 and CO, the 115 GHz J=1-0 transition of CO is what largely defines observed molecular clouds as H_2 is not as easily observed¹.

Large molecular clouds (≥ 100 pc in diameter) are host to complex substructure hierarchies with filaments (observed on the large scale with mid- to far-infrared observations) containing denser cores which theoretically undergo gravitational collapse creating single or binary stars (Kennicutt and Evans, 2012). The evolutionary stages associated with giant molecular clouds are classified by Kawamura et al. (2009) with types of observable cloud defined by stages of global massive star formation:

¹The abundance of H_2 can however be measured through the CO-to- H_2 conversion factor (Hennebelle and Falgarone, 2012) $X_{CO} = N(H_2)/W(CO)$

1. Type I: ‘starless’ molecular clouds without HII region signatures. While there can be stars present there aren’t any with high enough mass to ionize HII regions.
2. Type II: molecular clouds with observable HII regions
3. Type III: molecular clouds with both HII regions and young stellar clusters

Despite the obvious variation in molecular cloud characteristics, it would appear that they broadly follow the above evolutionary structure. As they transition from neutral shielded gas to a home for sets of increasingly massive and numerous stars, the UV radiation and stellar winds from within photo-evaporates the cloud material until only the stars remain.

Once a large molecular cloud is sufficiently massive, it can develop filamentary regions of compressed gas as a result of sub- and super-sonic turbulence (Larson, 1982). Within these densest regions, a developing protostellar core on an ideal evolutionary path follows four stages:

1. Core Assembly: Cores contract and increase in density as they collect gas, and expand under the influence of external shocks.
2. Collapse: Occurs once α_{cr} , a core’s critical virial parameter, is reached². Runaway collapse begins once a critical number density of $n \sim 10^6 \text{ cm}^{-3}$ is reached after which core density sharply increases.
3. Accretion: At this point a protostar or protostars have formed in the core. Turbulence in the cloud core increases as a result of outflows.
4. Dispersal: The protostar photoevaporates the cloud material from the inside.

In terms of these dense cores in which star formation occurs, according to simulations by Offner et al. (2025) star formation can be seen as quite inefficient with only $\sim 8\%$ of cores playing host to protostars and most cores dispersing before star formation can occur.

Molecular clouds do not undergo collapse naturally as there is a local internal pressure that balances the gravitational pressure, keeping the system in equilibrium in the absence of perturbation. The sources of the necessary gravitational instability to incite collapse are turbulent eddies within the gas that transfer kinetic energy from the large scale structure to the smaller-scale filaments, providing them with constant movement (Larson, 1981). Although it is not known exactly where this turbulence comes from, possible sources include the shockwave from a nearby supernova or ambipolar diffusion³ as a result of an interstellar magnetic field (Montmerle et al., 2006).

Lending further credence to the pathway to star formation via supernova influence, a series of magnetohydrodynamical simulations carried out by Iffrig and Hennebelle (2015) demonstrate that a supernova

²A key parameter determining whether molecular cloud cores are sub- or super-critical is the virial parameter $\alpha = \frac{5\sigma_v^2 R}{GM}$ with σ_v , M and R being the one-dimensional velocity dispersion, core mass and core radius respectively. Sub-critical cloud fragments aren’t gravitationally bound and can release their constituent gas and dust back into the surrounding ISM while super-critical fragments are gravitationally bound and susceptible to gravitational collapse under perturbation and hence can form stars (Kauffmann et al., 2013). The boundary between sub- and super-critical cores can be defined as a critical virial parameter $\alpha_{cr} = 2$ below which cores can be considered super-critical

³magnetostatically supported cores accreting neutral matter from the surrounding material onto the core surface (Dib et al., 2007)

located outside a molecular cloud can transfer a small fraction of its momentum to the cloud’s gas without significantly affecting the gas mass. This becomes increasingly relevant when discussing Bok or cometary globules in the following sections.

In terms of individual stars forming within these cores, once a protostellar core exceeds the Jeans mass - i.e. the cloud mass at which its gravitational pressure acting inwards exceeds its gas pressure acting outwards - the core will undergo a star-forming runaway collapse under its own gravity, with the Jeans mass expressible as

$$M_J = \frac{4\pi}{3}\rho R_J^3$$

(with ρ being cloud density, $R_J = \frac{1}{2}\lambda_J$ being radius of cloud region. λ_J refers to a critical radius at which the potential energies for internal thermal pressure and inward gravitational pressure are equal.)

1.1.2 Young Stars and the Formation of Disks

The fundamental theoretical framework for planet formation today, at its most basic, differs little from its inception in Immanuel Kant’s nebular hypothesis (Kant, 1755); that the Sun and its planets originate from the collapse of a rotating cloud of dust and gas into a rotating disk, with the dust grains within the disk aggregating over time into planetesimals. In this section, the disk phase is explored.

The first immediate stage in stellar evolution following core collapse is the protostar, a compact core of superhot gas surrounded by its protostellar envelope. The youngest observed protostar is estimated to be 10kyr old, and such observations confirm hypotheses of stellar outflows or jets being present (Montmerle et al., 2006), which eject matter from the developing system, shedding angular momentum and facilitating the accretion of material onto the protostellar core (Bally, 2016).

Larson (1969) found that protostars undergo a non-homologous collapse where a central core reaches stellar densities before most of the outer cloud can collapse very far from its original configuration. Hence the subsequent stellar evolution consists of the growth of the star in the protostellar centre surrounded by material undergoing infall and accreting onto the developing stellar surface. This phase is inextricably linked with the formation of circumstellar disks; Ahmad et al. (2024) find that the transition between protostar and inner disk-hosting system is smooth and continuous. As the protostar accretes material from this surrounding envelope its angular momentum increases to a point that it reaches breakup velocity. The resultant shedding of mass forms a closely-bound circumstellar disk which spreads outwards through excess angular momentum and accretion. This first disk holds the majority of the mass in the system, and if an outer disk is already present the two will merge as the inner disk spreads.

The aforementioned outer disk forms as a result of high angular momentum material surrounding the protostar being unable to fall directly onto the developing star and thus accumulating along the plane perpendicular to the protostellar core’s initial angular momentum (Galli and Shu, 1993). Initially the inner disk has no boundary with its central protostar and both are symbiotic, with a continuous fluid-like interface (Larson, 1969). The star accretes directly from the inner disk, and as the envelope is depleted of

material through accretion the system stabilises with a star-disk-envelope configuration. As the system develops, the disk flares with increasing radius (Kenyon and Hartmann, 1987) if the temperature profile throughout is between $T \propto r^0$ and $T \propto r^{-1}$ (see Chiang and Goldreich 1997), which is a condition most disks satisfy.

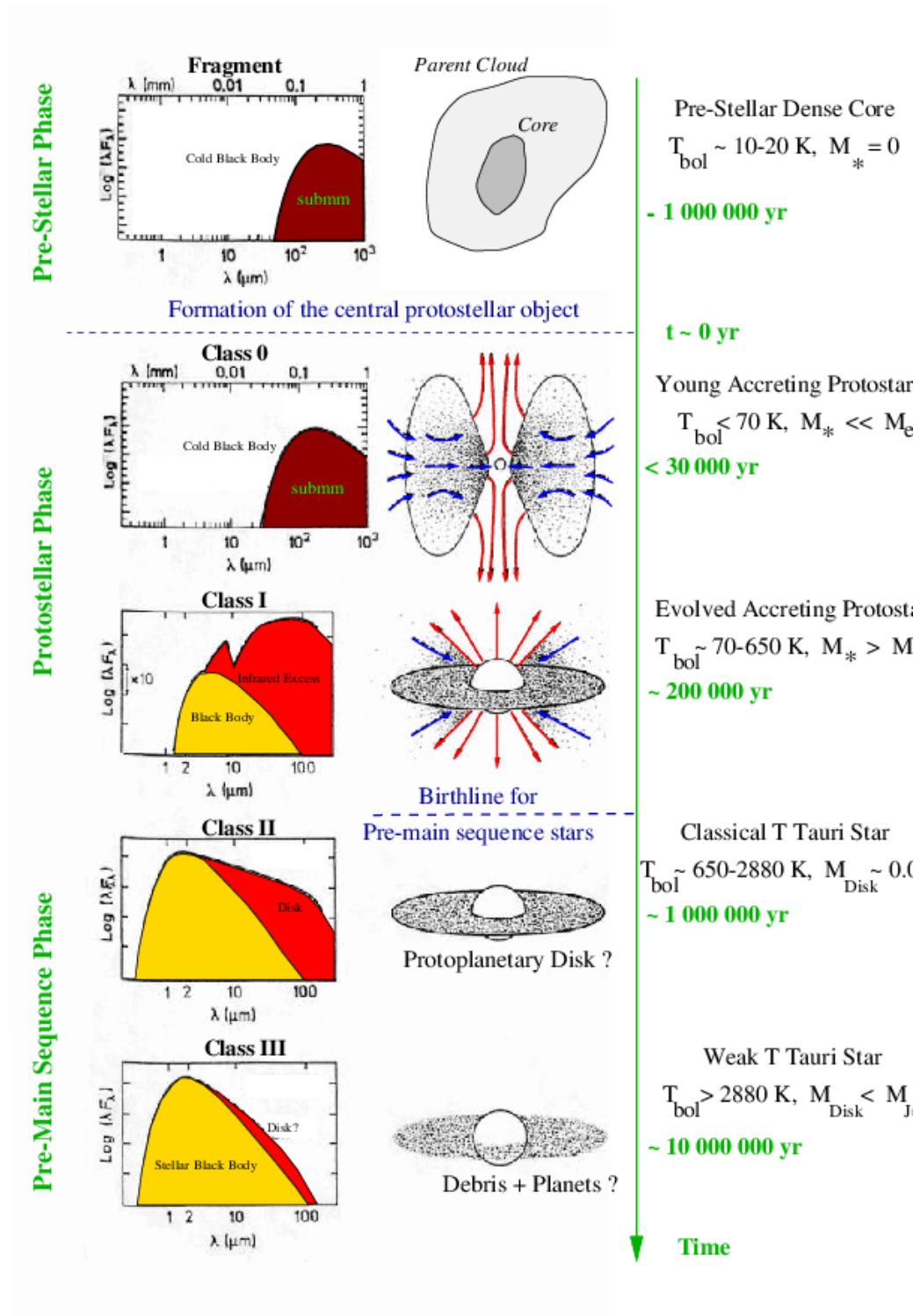


Figure 1: Figure demonstrating YSO evolution from André (2002).

The evolution of such a disk is categorized by class of central young stellar object (YSO) depending on excess of infrared radiation - which is analogous with evolutionary stage of YSOs (Gail and Hoppe, 2010). Class 0/I refer to young protostars with optically thick envelopes hindering direct optical or near-infrared observation. The spectral energy distributions (SEDs, see Figure 1) for these sources rise into the mid-infrared, indicating a non-stellar blackbody curve from the heating of circumstellar material. Class II/III YSOs are pre-main sequence stars with strongly detectable disks and SEDs with the stellar blackbody contribution and a mid-infrared excess. These are classical T Tauri stars (Joy, 1945), characterized by mid-infrared excess, variability in brightness and low luminosity among other attributes, around which most of the disks observed to date are found (see Figure 1).

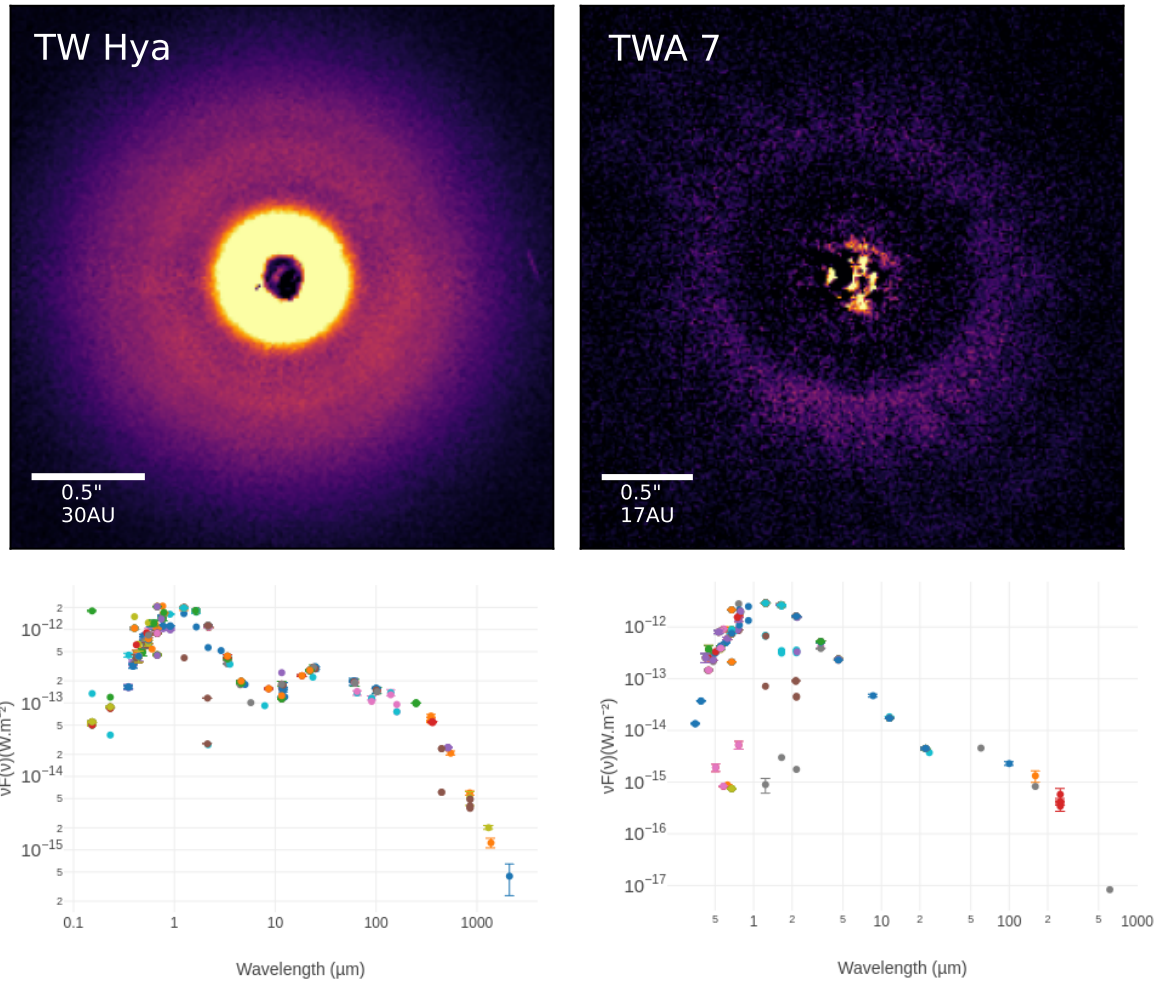


Figure 2: Stages in disk lifetime: TOP ROW: *H*-band SPHERE images. BOTTOM ROW: Spectral energy distributions. LEFT: *TW Hya*, an example of a gas- and dust-rich disk (class II). Despite having an age of 8 ± 3 Myr's (Sokal et al., 2018), *TW Hya* exhibits the characteristics of a young, massive disk. RIGHT: *TWA 7*, a debris disk (class III), aged 10 ± 2 Myr (Luhman, 2023).

Gas-rich protoplanetary circumstellar disks have an approximate average lifetime of ~ 8 Myr (Michel et al., 2021), which is assumed to vary with strength of the local far-ultraviolet (FUV) field (Johnstone et al. 1998, e.g. Concha-Ramírez et al. 2019) as FUV radiation truncates disk lifetimes through heating

of gas and dust, raising thermal pressures and creating a flow of material into the surrounding interstellar medium (Adams et al., 2004). Over the course of this disk lifetime, the gas is depleted through internal and external photoevaporation, the formation of massive planets and other processes - giving rise to two distinct stages in disk evolution (*e.g.* Wahhaj et al. 2010):

1. Gas- and dust-rich disks, see Figure 2, left.
2. Debris disks, in which gas has been depleted and disk material lies in a flat plane, much like the Kuiper Belt in our own Solar System (Matthews et al., 2014). See Figure 2, right.

As a general rule the separation between blackbody curves for the star and disk in the SED can be related to physical separation between star and disk material; the further away such material is, the cooler it will be, with a blackbody peak further into the mid-infrared. A ‘transition disk’ refers to a disk in an intermediate stage where there is a separation in the SED between the stellar blackbody curve and the disk blackbody curve, manifesting as a dip around $10\ \mu\text{m}$ in the SED (Strom et al., 1989), see Figure 2. The most extreme examples when seen in resolved images will sport a visible large empty cavity between inner and outer disk material (See for examples of transition disks Mayama et al. 2012, Pinilla et al. 2018).

1.2 Planet Formation

Within a disk there are two main channels by which planets are believed to form: core accretion and gravitational instability.

Under the core accretion model, small (μm scale) dust grains settle towards the circumstellar disk midplane and grow through inter-granular collisions. These aggregations of grains continue up in scale, with mm- to cm-sized grains aggregating under electrostatic and gravitational forces until large enough (km scale) to decouple from the disk gas (Lissauer, 1993). At this stage gravitational interactions dominate between developing planetesimals, with growth occurring through binary collisions until nascent protoplanets are large enough to be dynamically isolated from each other. At a sufficiently massive size planets can gravitationally accumulate gas, which can significantly contribute to the depletion of gas within a disk region, acting as a limit on how much gas can be accreted (Lissauer, 1993).

However, core accretion timescales are not sufficiently long to create massive (large enough to be directly imaged, *e.g.* brown dwarves and massive super-Jupiters) planets within the disk lifetime (*e.g.* Emsenhuber et al. 2021). Such objects are theorized to be produced through gravitational instability causing the disk to fragment into bound objects orbiting the central star which can undergo collapse into high mass planets and brown dwarves (Boss, 1997).

This gravitational instability model suggests that this fragmentation can occur within a sufficiently cool, massive and rotational shear-supported disk. Such conditions are described by the Toomre Q parameter (Toomre, 1964)

$$Q = \frac{c_s \Omega}{\pi G \Sigma} < 1$$

with c_s being disk sound speed (which is proportional to temperature), Ω being angular velocity, G being the universal gravitational constant and Σ being the disk surface density. As Q approaches 1 the disk becomes self-gravitating and susceptible to gravitational collapse.

This process eschews any need for a solid planetesimal core in favour of direct collapse of gas under self-gravity. However it would appear that such a mechanism is only capable of forming exceptionally high-mass bodies at very large initial separations. Rafikov (2005) finds that for gravitational instability to form giant planets at 10AU local temperature must be in excess of 10^3 K, disk mass must exceed $0.7 M_\odot$ and disk luminosity must exceed $40 L_\odot$, and the required mass of an object formed at a large distance such as 100 AU sits at $\sim 10 M_{\text{Jup}}$. In fact, Kratter et al. (2010) argue that for a body forming at a wide separation from an A star, the masses of gravitational-instability-driven fragments exceed the deuterium burning limit for planets, and hence only substellar objects can form through gravitational instability.

2 Observations of Planet Formation

The earliest studies of circumstellar disks were generally limited to conclusions drawn from the SEDs and photometric excesses of candidate YSOs. However, the ongoing development of instrumentation has provided a host of observational techniques for probing many aspects of disk structure and planets within.

This project focuses primarily on polarimetric imaging of starlight scattered off the small grains constituting the disk surface, and radio interferometry as a method of probing the large dust grain midplane of disks and the disk gas content. In the sections below, both techniques are described in detail.

2.1 Scattered Light Observations

2.1.1 Near-Infrared Observations

Within the field of disk study, near-infrared (NIR) observations are primarily used to trace the surface layers of disks, as this wavelength regime is sensitive to the μm -scale dust particles that make up the upper layers of a disk and thus can be used to analyse disk geometry and grain size distribution. Characterising the behaviour of light incident on these particles under Mie scattering (Mie, 1908), it is understood that the fraction of light scattered by a single dust grain is given by the scattering albedo ω ; for particles smaller than the wavelength of incident light most of the light is absorbed by the particle. As the particle size increases, so too does the single scattering albedo. In this regard - to massively simplify the relationship between particle size and wavelength of light - for a micron-sized particle, wavelengths in the micron regime will trace the light scattered by these particles (van de Hulst 1957, Bohren and Huffman 1983). Hence the near-infrared regime (1-5 μm) is ideal for tracing disks in scattered light.

In order to successfully capture the disk morphology in scattered light, it is necessary to strip the starlight from the total signal as scattered light constitutes only a small fraction of the total light emitted by the system in this regime. The preferred method used by this project is polarimetric differential imaging (PDI) (Kuhn et al., 2001), which can differentiate between partially linearly-polarized scattered light and largely unpolarized direct starlight. How this is used to obtain scattered light disk images is expanded on in the following section.

2.1.2 SPHERE Polarimetric Imaging

SPHERE (Spectro-Polarimetric High-contrast Exoplanet REsearch instrument) is an extreme adaptive optics system within the VLT (Very Large Telescope) that has pushed the limits of high-contrast disk imaging and proximity of observable structures to their host star (Beuzit et al., 2019). Benefitting from its SAXO system (SPHERE AO for eXoplanet Observation) that provides wavefront correction with a high-order deformable mirror, a fast Shack-Hartmann wavefront sensor operating up to 1.2 kHz and a

tilt-tip mirror for rapid correction of atmospheric turbulence and telescope vibrations (Fusco et al., 2016), SPHERE consistently achieves Strehl ratios⁴ of 0.9 for bright targets ($R \approx 9 - 10$ mag), which ensures diffraction-limited performance sufficient for resolving disk structure and in some cases large planetary companions (Beuzit et al., 2019). Incorporated into SPHERE is IRDIS (Infra-Red Dual-band Imager and Spectrograph) which provides dual-band imaging and polarimetry (Dohlen et al., 2008), with which all scattered light images used in this project are acquired.

In a SPHERE image of a circumstellar disk, the central star is masked out by a coronagraph to prevent the stellar light from obscuring the disk signal. What is seen after coronagraphic masking is a combination of stellar light and disk signal. After polarization differential imaging, the image becomes a depiction of the disk scattering surface in small dust grains, on which the angle of incidence of light has a strong forward- or back-scattering effect. The degree to which this is seen depends on the inclination of the disk to the observer, as can be seen in Figure 3.

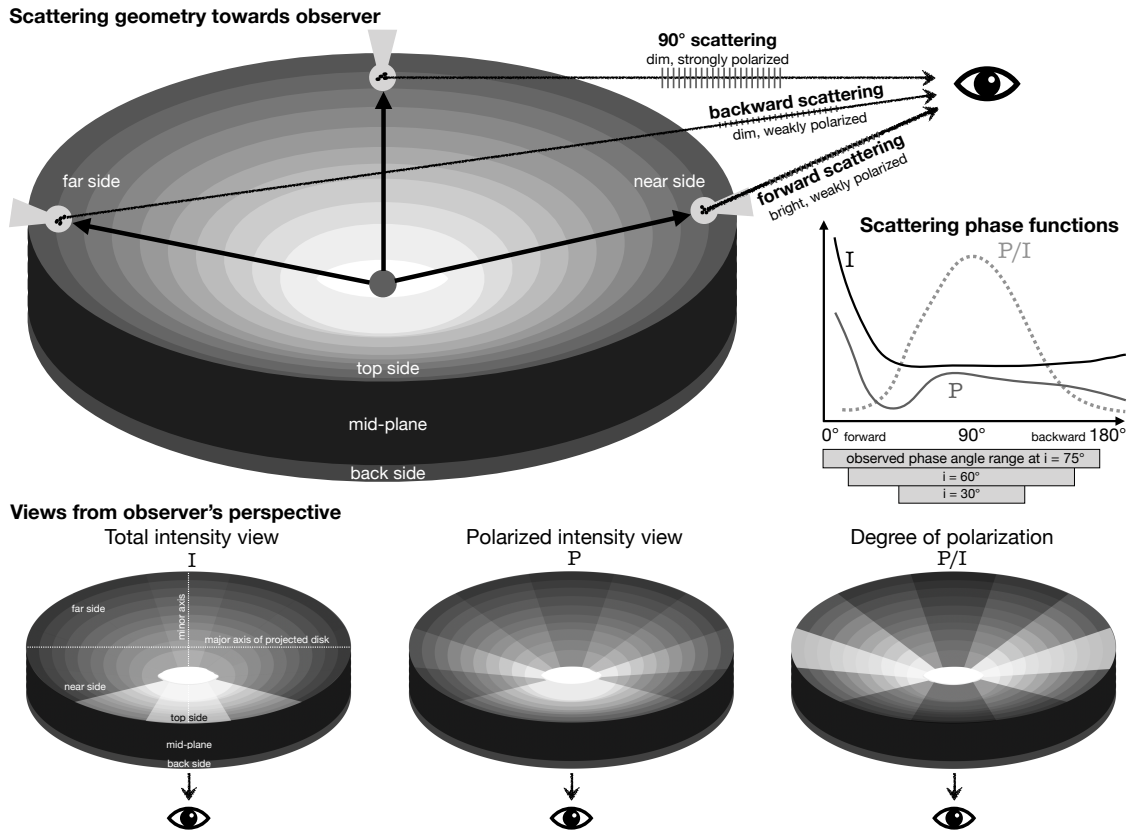


Figure 3: Figure 2 from Benisty et al. (2023) demonstrating the appearance of disks under PDI. TOP: Scattering locations for forward, backward and 90° scattering. BOTTOM LEFT: Total intensity of disk signal if polarization effects not considered. BOTTOM MIDDLE: Polarized intensity, i.e. what an observer sees in a SPHERE image. BOTTOM RIGHT: the degree of polarization.

The intensity or brightness of disk signal in an image is dependent on many effects. Firstly the

⁴Strehl ratio is defined as the ratio of peak intensity of an observed point spread function (PSF) to that of an ideal diffraction-limited PSF. Values close to 1 indicate near-diffraction-limited performance

disk signal intensity as a function of radial distance from central star is unsurprisingly governed by the inverse-square law so that $I(r) \propto 1/r^2$, as the intensity of starlight itself drops off with distance from the stellar point source. The polarization intensity, the next most important intensity parameter for SPHERE images is influenced by the angle-dependent nature of polarization; linear polarization peaks at scattering angles near 90° and drops off at smaller (forward-scattering) and larger (back-scattering) angles. With disks inclined to an observer, the flared disk surface will follow an intensity profile represented in Figure 3, where regions along the disk major axis about which it is inclined will be seen as brightest due to the near- 90° scattering angles (hence the bright fringes seen in scattered light disk images, hereafter referred to as the ‘ansae’), and the disk near-side will be brighter than its far-side as forward-scattering will provide a higher degree of linear polarization than backward scattering (Benisty et al., 2023).

SPHERE observations consist of ‘polarimetric cycles’ (de Boer et al., 2020) over which the internal half-wave plate (HWP) is modulated to provide the U_+ , U_- , Q_+ and Q_- terms of the Stokes Vector (Stokes, 1851):

$$\mathbf{S} = \begin{bmatrix} I \\ Q \\ U \\ V \end{bmatrix}, \quad (1)$$

where I is total light-beam intensity, Q and U are linearly polarized light contributions (in which the disk signal can most prominently be seen) and V refers to circularly polarized light. The HWP angles 0° and 45° provide the Stokes Q images and angles 22.5° and 67.5° provide the Stokes U images (van Holstein et al., 2020).

For symmetric scattering patterns (such as those expected from circumstellar disks), the polarization is re-expressed in a co-ordinate system aligned with the azimuthal direction from the central source (de Boer et al., 2020) as follows:

$$Q_\phi = -Q \cos(2\phi) - U \sin(2\phi)$$

$$U_\phi = Q \sin(2\phi) - U \cos(2\phi)$$

with ϕ being the azimuthal angle in the image plane. All scattered light images used in this project are Q_ϕ images, in which all the azimuthally polarized signal is represented as positive pixel values and all radially polarized signal as negative values (de Boer et al., 2020)⁵.

2.2 ALMA and Millimetre Emission

Infrared imaging is valuable for probing the upper disk layers but cannot trace the total material content of disks. In contrast (sub-)mm emission is a valuable tracer for the total amount of material in disks, as

⁵Note however that in this thesis, in any case where a colourbar is present to demonstrate flux values, pixel values have been converted to flux units

μm -scale dust grains are optically thin in the (sub-)mm regime and longer wavelengths probe larger dust grains in cooler disk regions which contain the majority of the dust mass (Wyatt and Dent 2002, Draine 2006). For instance from Hildebrand (1983) the total disk dust mass can be obtained with integrated flux from mm dust emission; with an assumption that the dust midplane is optically thin, isothermal and homogeneous in material, and with the radiation emitted at this wavelength regime being primarily due to thermal emission from solid particles, this mm flux is defined as

$$F_\nu = \kappa(\nu)M_d B_\nu(T_d)D^{-2}$$

with $\kappa(\nu)$, M_d and T_d being dust opacity, total mass and temperature respectively, B_ν being blackbody intensity and D being distance.

ALMA (Atacama Large Millimetre-submillimetre Array) is a state-of-the-art array consisting of 66 antennae that operate in configurations to simulate larger radio telescope dishes through interferometry. From diffraction theory, a single telescope with dish diameter D has an effective resolution $\sim \lambda/D$ radians. For longer wavelengths, resolution will degrade unless dish size is increased to compensate, for instance if looking to reach an angular resolution of 0.13 arcseconds at $\lambda = 1$ mm, it is necessary to use a 2 km diameter dish. Each antenna in an ALMA configuration acts as a boundary point for a simulated dish with a very large diameter, with the effective D being given by the length of projected baselines between pairs of antennae⁶. Thus resolution is then defined as $\sim \lambda/B_{\text{max}}$ with B_{max} referring to maximum distance between two antennae.

To briefly describe the interferometric setup of ALMA, a configuration of N antennae measures the interference pattern produced by $N(N-1)/2$ independent pairs of apertures in a basic 2-slit experiment, producing an interference pattern. Each antenna pair provides visibilities⁷ in the (u, v) Fourier image plane, one visibility for each integration time, channel, projected baseline and correlation.

In terms of the detection of disk gas content, the wavelengths at which molecular gas is observed are dependent on the molecular transitions of the target molecule, with this thesis primarily focusing on the (J=2-1) transition for ^{12}CO gas; the energy of such a transition is $E = 1/2\mu r^2\omega^2$ with $\mu = mM/(m+M)$ (m, M being the masses of the C and O atoms), and when corrected for the quantization of angular momentum in quantum mechanics $E = (h/2\pi)^2 J(J+1)/2\mu r^2$ - hence the associated wavelength for ^{12}CO (J=2-1) is 1.3 mm.

ALMA's view of ^{12}CO gas line emission involves centering a wide-range spectral window on the Doppler-shifted ^{12}CO (J=2-1) line with this range separated into frequency channels either side of the central line so that red- and blue-shifted gas emission is also tabulated. Over this frequency range the Doppler-shift of each channel can provide the velocity of cloud components towards and away from the observer. In the case of disks this can show outflows, infall and primarily the axis of rotation of disk material as one side of the disk will be redshifted in accordance with its rotation in the direction away from the observer and vice versa.

⁶Projected baselines refers to inter-antenna distance corrected for projection effects from light arriving earlier at one antenna when both antennae are inclined towards a source

⁷The visibility function is effectively the Fourier transform of the sky brightness distribution

For examples of ALMA observations of circumstellar disks, see ALMA Partnership et al. (2015) for resolved images of the spatially-resolved disk around HL Tau in gas line emission and dust continuum emission, in which detail was resolved down to a few AU. See also Andrews et al. (2018) for an ALMA survey of disks, where substructures in continuum emission such as rings, gaps and spirals were spatially-resolved down to similar scales.

3 The Gum Nebula

To set the scene, the Gum Nebula in which AT Pyx, the star of interest, is located is a HII region in the southern Milky Way, identified as an intermediate structure between classical HII region and typical supernova remnant by Chanot and Sivan (1983), located about 350pc away and with an angular size of $\sim 40^\circ$. AT Pyx is located within CG22, one of the Gum Nebula’s characteristic cometary globules, explored in the following section.

3.1 Cometary Globules

Returning to the area of star formation within molecular clouds, this project has a distinct focus on stars forming within the cometary globule association of the Gum Nebula. These formations as defined in Hawarden and Brand (1976) are distinctly comet-shaped, with compact, dusty, opaque heads and trailing tails. Such globules are classified as precursors (Reipurth, 1983) to Bok globules (Bok and Reilly, 1947), *i.e.* isolated dense clouds with masses on the order of 10 to 100 $M_\odot$ and sizes on the order of 0.1-1pc. Such globules may form one to a few isolated low-mass stars similar to the Sun (Elmegreen, 1998).

Cometary globules are suggested to have developed from the destruction of clumps of molecular cloud under extreme ultraviolet (EUV) and far ultraviolet (FUV) radiation from massive stars. EUV radiation can ionize and heat the low-density gas surrounding these clumps in a process referred to as ‘radiation-driven implosion’; The ionisation front, *i.e.* the final globule head, drives a shock into the initial cloud with a maximum velocity of the local soundspeed within the ionised gas. The gas at the ionisation front is highly compressed as the ionisation front pushes the shocked gas onto the axis of symmetry of the parent cloud. In this regard star formation can be triggered as this compression of gas can cause otherwise gravitationally stable clouds to become supercritical and subject to collapse (Bertoldi, 1989).

According to the models of Nakatani and Yoshida (2019), the molecular cloud clumps under EUV remain shock-compressed for the first 4kyr, with about half the initial cloud mass lost to photoevaporation over this period. Subsequent mass-loss rates depend on the initial clump surface area. This would appear to explain the appearance and distribution of the globules present in the Gum Nebula, an association of at least 32 such globules with tails pointing generally radially outwards from the centre of the Vela OB association (see Choudhury and Bhatt 2009). The central ionising stars are assumed in the literature to be ζ Pup, γ^2 Vel - the two brightest stars in the Vela OB association - and Vela XYZ, the progenitor of the Vela supernova remnant also located within the Gum nebula (See Kim et al. 2005, Choudhury and Bhatt 2009, Yep and White 2020).

3.2 AT Pyx

AT Pyx (IRAS 08267-3336, WRAY 15-220) is known to be embedded in the head of CG22 (Sahu and Sahu, 1992), and has a most recent distance estimate of 370 ± 5 pc from Gaia DR3 (Gaia Collaboration

et al., 2021). It was first identified as a pre-main sequence star by Pettersson (1987), with high $H\alpha$ emission. Its SED, assembled from various photometric catalogs (shown in Figure 4) has a clear infrared excess and a dip in emission at $\sim 10\mu\text{m}$, typical for a transition disk with an inner cavity. It has strong V-band variability, found with an amplitude of 1.45mag and timescales on the order of days (ASAS-SN catalog, Jayasinghe et al. 2019), which could be interpreted as AT Pyx’s inner disk material periodically obscuring the star.

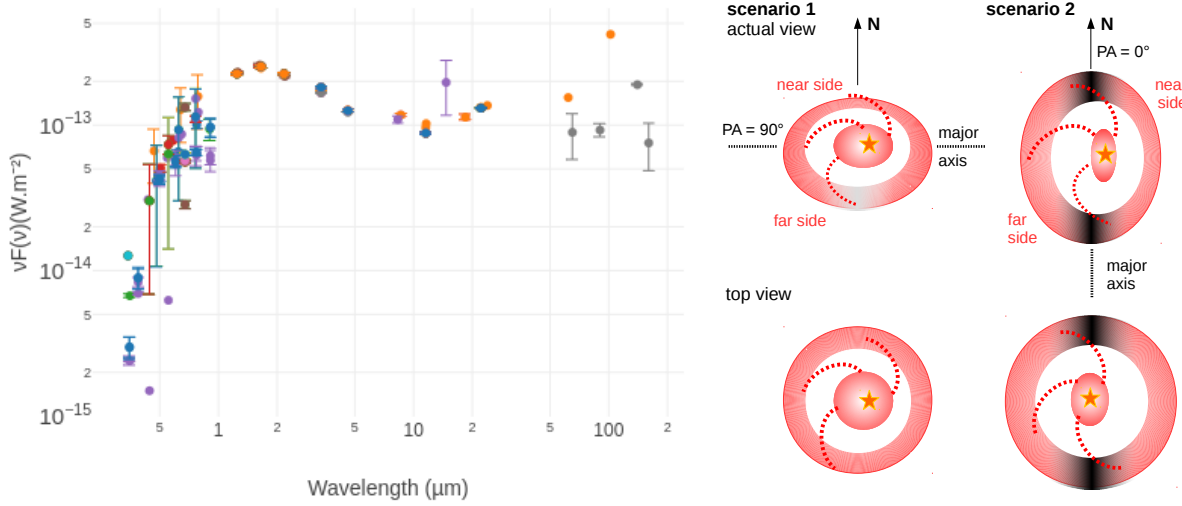


Figure 4: LEFT: SED for AT Pyx from Vizier (Ochsenbein et al., 2000). RIGHT: Figure 4 from Ginski et al. (2022) showing the two disk orientation scenarios determined.

Ginski et al. (2022) presented SPHERE and NACO (NAOS-CONICA - Nasmyth Adaptive Optics System (NAOS) and Near-Infrared Imager and Spectrograph (CONICA), Lenzen et al. 2003, Rousset et al. 2003) observations of its disk in scattered light in which its extent was measured to be $\sim 126\text{AU}$, and two scenarios were presented for on-sky orientation. Scenario 1 has the major axis congruent to the horizontal axis in the image plane, which marks the disk as possibly eccentric as the fitting of an ellipse to the disk ansae gives an ellipse centre 55 mas offset from the star position - an offset that cannot be explained through projection effects. The second scenario allows for a circular disk with the disk major axis at an angle of 90° to the horizontal axis in the image plane (see Figure 4). Hence, one of the first tasks for this project was to accurately characterise the disk’s orientation, as described in Section 5.1. Three spiral features were identified and tentatively investigated through comparison to simulation results to provide a rough perturbing planet mass of ~ 3 Jupiter masses.

Now with a wider range of data, this project aims to draw further conclusions on this system and investigate a number of scenarios driving the unique morphology and characteristics of AT Pyx’s disk. The following sections detail all methodology and findings for this endeavour.

4 Observational Data

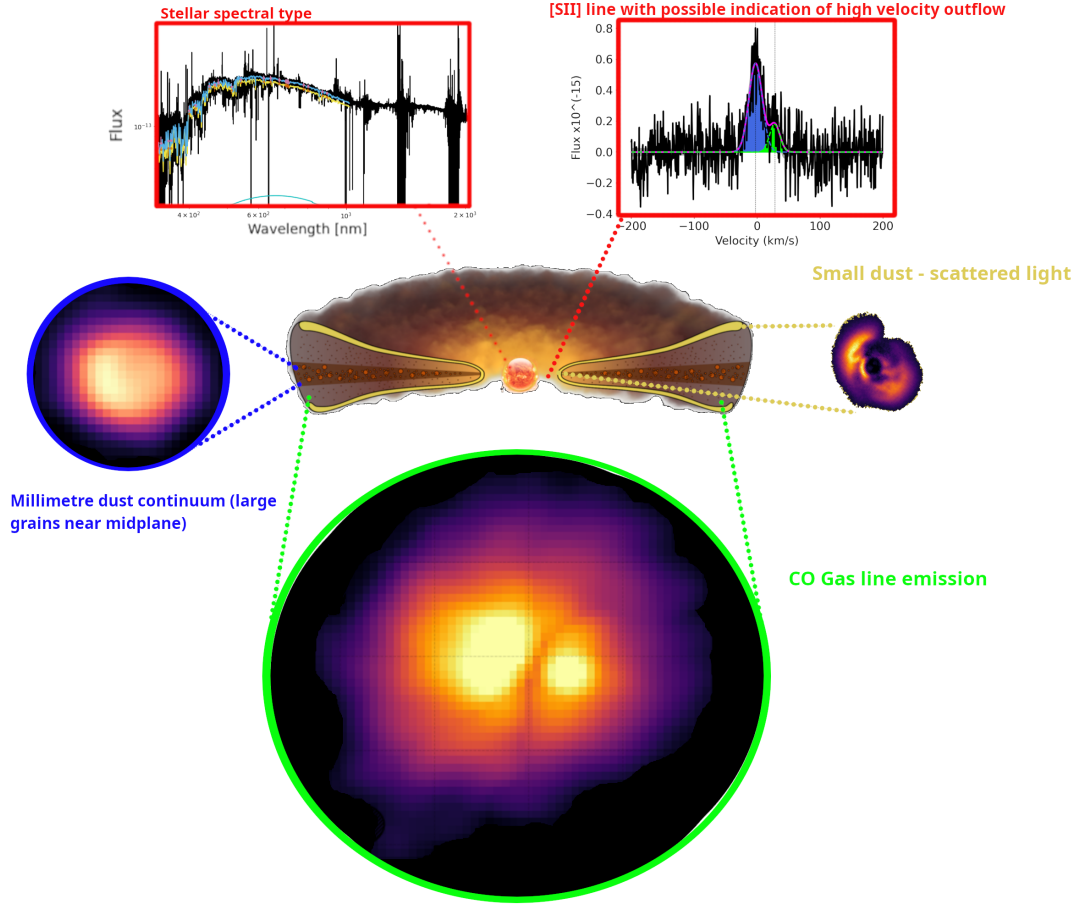


Figure 5: Composite of all data on AT Pyx used in this project, with corresponding disk components. All images are to scale with one another.

AT Pyx was observed with ESO’s VLT instruments XSHOOTER (see Section 4.1), SPHERE (see Section 4.3) and ESPRESSO (see Section 4.1) as well as ALMA (see Section 4.2), providing a comprehensive overview of stellar and disk properties.

The spectroscopic data available from XSHOOTER and ESPRESSO provides views of the most compact regions of the disk as they deal with stellar parameters and can trace behaviour of the central star as well as the levels of internal and external photoevaporation for the disk. The resolved image data from SPHERE and ALMA meanwhile provide large-scale depictions of the disk’s top scattering surface, dust midplane and gas content, from which the suite of analytical tools used in this project can derive conclusions about the processes at play within the disk. This overall coverage of the disk from the data selection is visualised in Figure 5.

Table 1: XSHOOTER Dataproducts. All runs taken on 30/10/2023. Seeing varied between 2.51 and 3.31 which is poor but can be compensated for by varying slit width to accept more light - in the case of these observations spectral resolution would appear to have remained adequate for the purposes of the parameters calculated using this data. The pipeline for reduction of XSHOOTER data makes use of the Molecfit tool (Kausch et al., 2015) to remove telluric lines resulting from absorption by atmospheric water molecules and on the night of observation precipitable water vapour levels remained at ~ 2.75 mm, well within the threshold for Molecfit correction.

Wavelength (nm)	Resolution ($\frac{\lambda}{\delta\lambda}$)	Observation Time	Integration Time Per Pixel (s)	Slit Width (arc-sec)
298.920 - 555.980	1512	07:42:01.096	90	5
994.020 - 2478.960	1029	07:42:08.926	80	5
298.920 - 555.980	1040	07:42:01.096	90	5
533.660 - 1020.000	18340	07:44:31.336	640	1
298.920 - 555.980	5453	07:44:26.166	1000	0.4
994.020 - 2478.960	11424	07:44:34.423	1000	0.4

4.1 Spectroscopic Data - XSHOOTER and ESPRESSO

XSHOOTER is a multi-wavelength spectrograph with a spectral range of 300-2480 nm over three sub-ranges as obtained with its three echelle spectroscopic arms: ultraviolet 300-559.5 nm, visible 559.5-1024 nm and infrared 1024-2480 nm (Vernet et al., 2011). The dataproducts from XSHOOTER observations of AT Pyx are shown in Table 1. This provides intermediate-resolution spectra for AT Pyx’s stellar type, see Figure 6.

XSHOOTER data (ESO Program ID: 112.25BZ.001) and interpretations were provided by Claes et al. (in prep) who used the methodologies described in Manara et al. (2023) to determine spectral type, effective temperature, luminosity, mass and accretion rate. Two values for mass and accretion rate are present from using two different stellar evolution models, Siess et al. (2000) (SIESS) and Baraffe et al. (2015) (B15) which provides a mass range 1.2-1.3 $M_{\odot}$ and accretion rate range 3.0093 - 3.4312×10^{-9} $M_{\odot}/\text{yr}$. Distance is already known from GAIA Data Release 3 (Gaia Collaboration et al., 2021) but included here among other ‘fundamental’ stellar parameters for the sake of completeness. See Table 2 for all parameters obtained from XSHOOTER data.

Table 2: *Stellar properties obtained from XSHOOTER spectrum of AT Pyx.*

Epoch	1	2
Distance (pc)	370	370
Spectral Type	K2.5	K2.5
T_{eff} (K)	4625	4625
Luminosity ($L_{\odot}$)	1.098	0.922
Mass - B15 ($M_{\odot}$)	1.278	1.221
Accretion Rate - B15 ($M_{\odot}/\text{yr}$)	3.4312×10^{-9}	3.0737×10^{-9}
Mass - SIESS ($M_{\odot}$)	1.317	1.247
Accretion Rate - SIESS ($M_{\odot}/\text{yr}$)	3.3288×10^{-9}	3.0093×10^{-9}

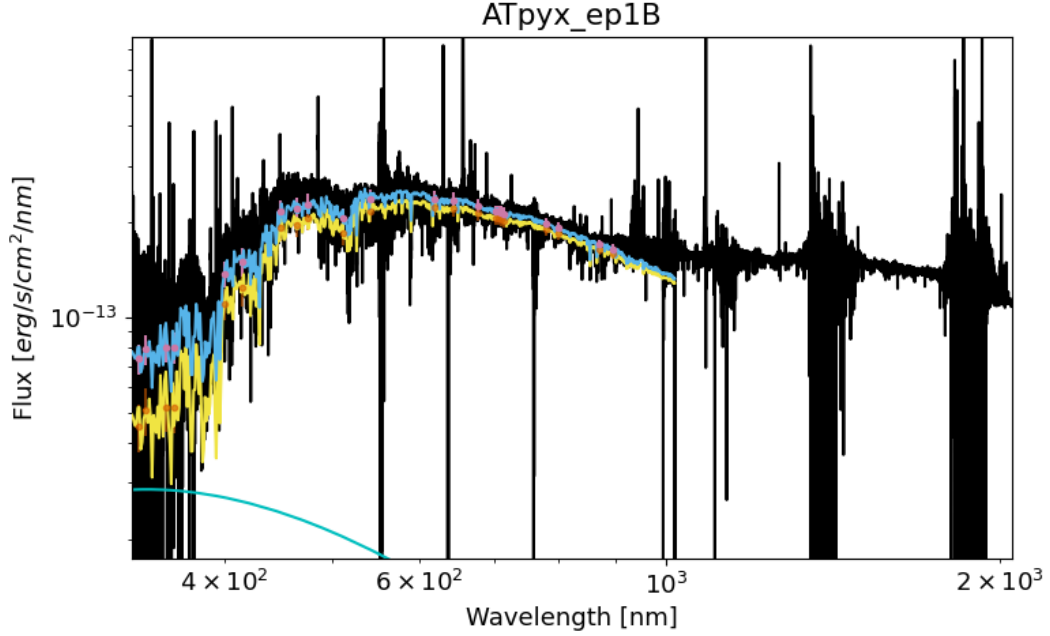


Figure 6: *Characteristic spectrum for AT Pyx from second epoch of XSHOOTER observations.*

ESPRESSO is another VLT echelle spectrograph operating at a high resolution with a spectral range 380-788 nm with the aim of detecting Earth-like exoplanets (Pepe et al., 2021). ESPRESSO data and related interpretations on AT Pyx were obtained from a yet-unpublished project by Campbell-White et al. (ESO program ID 111.24UB.001) and are summarised here. It was seen that the low-velocity components of the spectra are blueshifted, indicative of a disk wind which can ‘loosen’ material from the disk surface to allow it to infall and accrete onto the stellar surface. Additionally in the [SII] line fit this group found tentative evidence for a redshifted component (see Figure 7) which could indicate the presence of a high-velocity outflow.

It should also be noted that no strong [NeII] lines were found by this research group. [NeII] lines are

a strong indicator of external photo-evaporation (Pascucci et al. 2007, Alexander 2008), and so it can be assumed that AT Pyx's disk is minimally or not at all affected by external UV radiation. In Section 7.4 the assumed FUV flux at the location of AT Pyx is calculated.

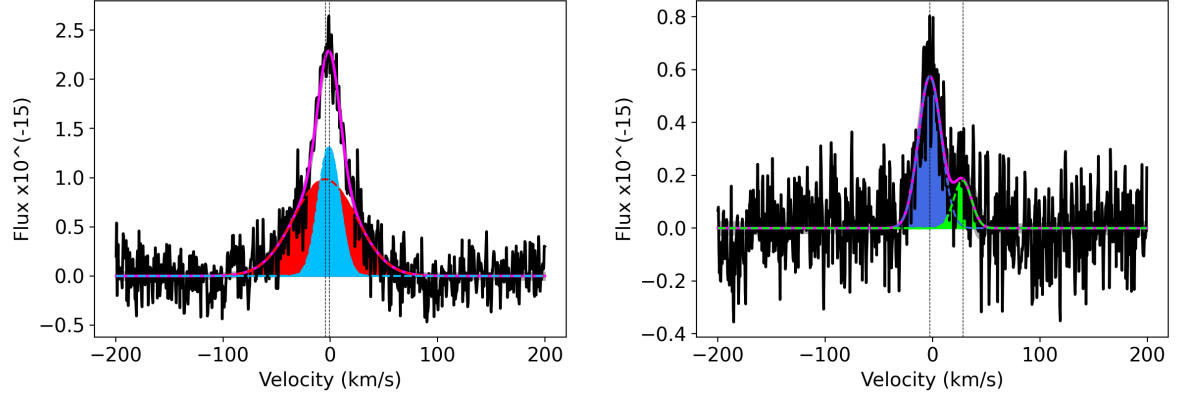


Figure 7: *ESPRESSO Spectra.* LEFT: *[OI] fit.* RIGHT: *[SII] fit.*

Unfortunately no data is available on the observing conditions for these ESPRESSO observations.

4.2 Gas Line and Continuum Data - ALMA

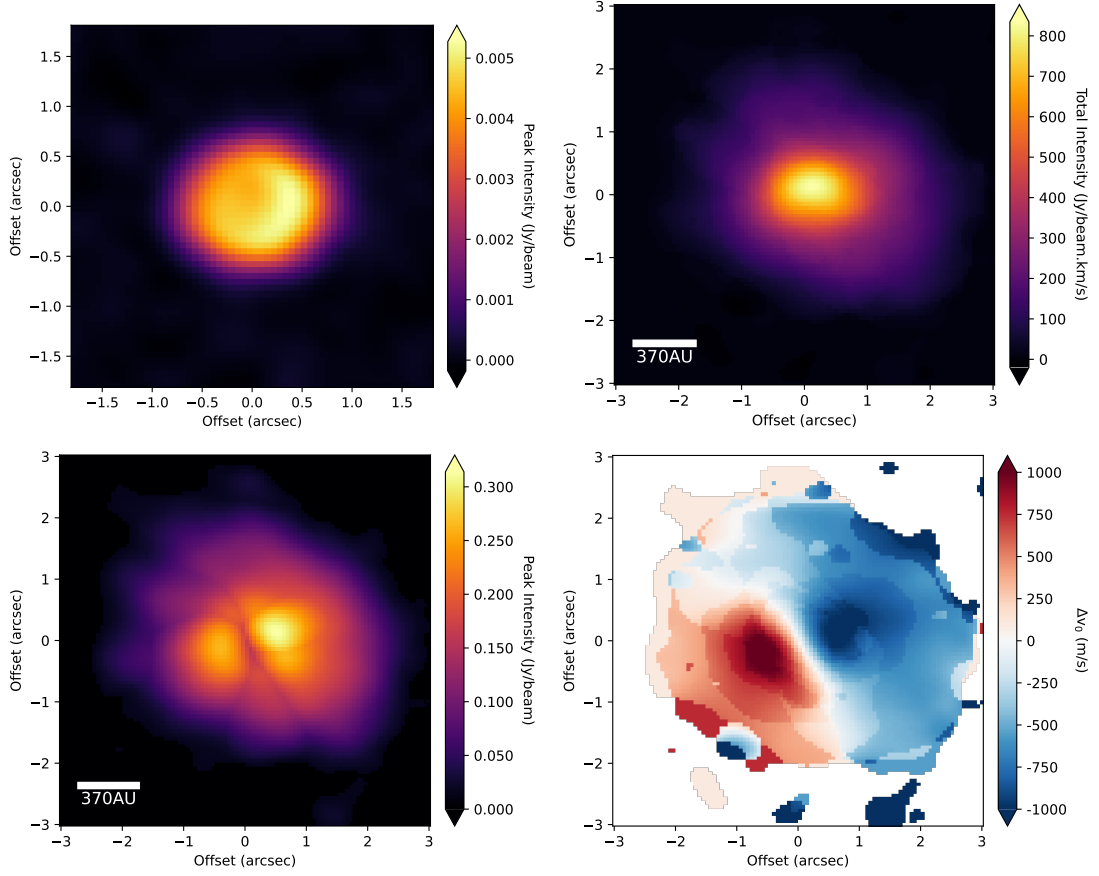


Figure 8: TOP LEFT: *ALMA 1.3 mm dust continuum image.* TOP RIGHT: *ALMA line emission moment 0 image - gas line emission total intensity.* BOTTOM LEFT: *ALMA line emission moment 8 image - gas line emission peak intensity.* BOTTOM RIGHT: *ALMA line emission moment 1 image - intensity-weighted velocity map.*

AT Pyx’s disk was imaged with ALMA at Band 6 (230GHz) in the molecular line $^{12}\text{CO}(J=2-1)$ - providing a datacube of 3838 spectral slices (13 of which cover the gas emission at a velocity resolution of 320 m/s) - and in 1.3 mm dust continuum emission with beam size for both datasets being $539 \times 397 \text{ mas}^2$ after reprocessing. This observation is part of a DESTINYs (Disk Evolution Study Through Imaging of Nearby Young Stars, Ginski et al. 2020) survey intent on investigating the relationship between disk dust-mass and formation of substructures through combination of scattered light and mm dust continuum observations (ALMA Project ID: 2021.1.01705.S). The re-reduction of raw ALMA data was carried out by Dr. Anibal Sierra. See Table 3 for image and cube parameters.

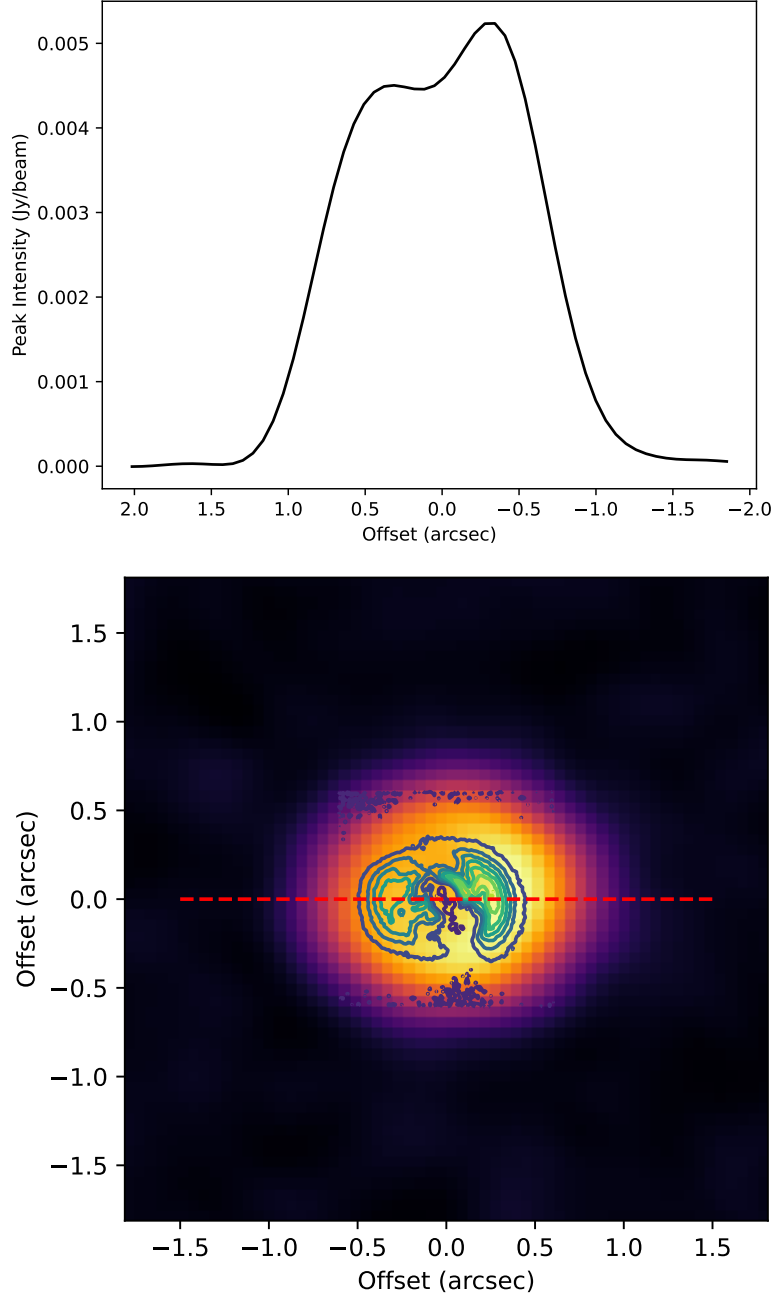


Figure 9: TOP: Radial intensity profile of continuum image. BOTTOM: Scattered light image contours overplotted on continuum image. Red line indicates radial cut over which intensity measurements are taken.

The resolved dust continuum image provides tentative evidence for a cavity in the central region (see Figures 8, top left, 9), and when overlaying the SPHERE scattered light H-band image on the continuum image it is notable that the bright arcs seen in the east and west of the disk appear to trace the cavity region. Continuum emission extends out to $\sim 1''$ (370 AU), half as extended as the peak intensity ^{12}CO emission ($\sim 2''$, 740 AU).

From the mm flux obtained from aperture photometry of the dust continuum image, disk dust mass

Table 3: ALMA image parameters

Data	Robust	Beam Size (arcsec)	rms (mJy beam ⁻¹)	Vel resolution (m s ⁻¹)
Continuum	0.5	0.539×0.397	4.2	-
¹² CO (J=2-1)	0.5	0.539×0.397	4.2	320

can be calculated using the following relation (Hildebrand, 1983):

$$M_d = \frac{F_\nu D^2}{B_\nu(T) \kappa_\nu}$$

with F_ν being integrated dust continuum flux, D being distance to the system, and the Planck function expressible in the Rayleigh-Jeans form $B_\nu \approx 2\nu^2 kT/c^2$ in mm wavelengths (Williams and Cieza, 2011). Disk temperature can be found to be $T = 43(L/L_\odot)^{0.25} \approx 43$ K (Kristensen et al., 2012) as AT Pyx was found to have a luminosity of $\sim 1 L_\odot$ from the XSHOOTER observations. Dust opacity

$$\kappa_\nu = 2.3 \left(\frac{\nu}{230\text{GHz}} \right)^\beta \text{ cm}^2\text{g}^{-1}$$

varies with grain size distribution as represented by dust spectral index $\beta \approx 0 - 2$ (Hartmann 2008, Andrews et al. 2013, see also Sheehan et al. 2022). In this case $\beta = 1$ is used giving a dust opacity of $2.3 \text{ cm}^2\text{g}^{-1}$. A system distance of 370 pc is obtained from Gaia DR3 (Gaia Collaboration et al., 2021).

Integrated flux density is obtained using aperture photometry of the dust continuum image to sum up all the pixel flux density values within the region of continuum emission. The summated values are then converted from Jy/beam to Jy by dividing this figure by beam volume $\Omega_b = \frac{\pi}{4 \ln 2} \times b_{maj} \times b_{min}$ with beam major and minor axes in units of pixels. Final integrated flux density stands at $F_\nu = 24.88$ mJy, thus the disk dust mass is calculated to be $\sim 97 M_\oplus$. Assuming a 1:100 relationship between dust and gas mass in circumstellar disks yields a total disk mass of $\sim 9700 M_\oplus$ or $\sim 30.5 M_{\text{Jup}}$.

From moment 1 maps of the gas line data it can be seen that the disk major axis (perpendicular to the dividing line between red- and blue-shifted sections) lies angled slightly against the image x-axis (west-east on-sky) (see Figure 8). This validates scenario 1 for the disk’s on-sky orientation from Ginski et al. (2022) (see Section 3.2).

There is a notable brightness asymmetry in both gas line and dust continuum emission, and a darker region in the centre of the moment 8/peak intensity map which could possibly be related to the cavity in continuum emission or due to the absorption of light from globule cloud material at the systemic velocity (see Figure 13). Further analysis of the ALMA line data is detailed in Section 5.1.

4.3 Infrared/Scattered Light Data - SPHERE

VLT’s SPHERE instrument is an adaptive optics and coronagraphic system designed for acquisition of data on extra-solar planetary systems and as such is ideal for imaging nearby circumstellar disks in the

near-infrared. Images of AT Pyx's scattering surface were obtained using the IRDIS instrument (Dohlen et al., 2008) - a dual-band imager with a spectral range of 0.95 - 2.32 μm - in 2017 and 2024. Observations were made in polarimetric mode with IRDIS' H-band filter (centred at 1.65 μm with a 300 nm bandwidth) as most disks tend to be brightest in the H-band (Avenhaus et al., 2018).

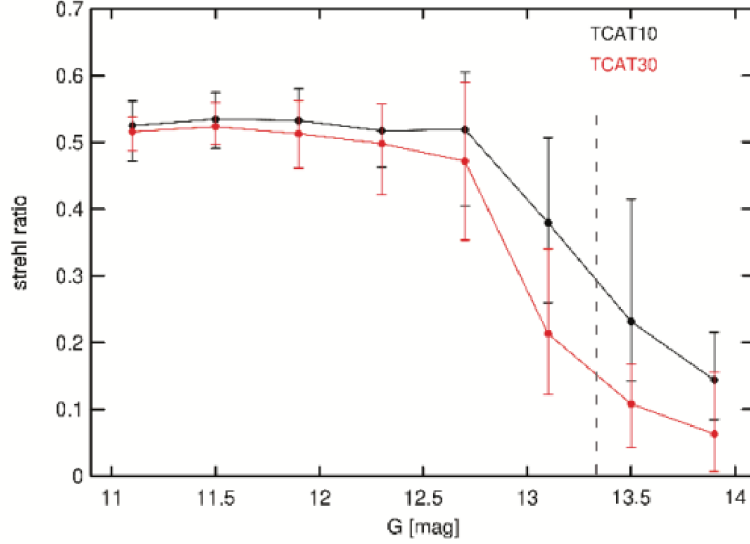


Figure 10: *SPHERE* strehl ratio as a function of *G* magnitude, taken from the *SPHERE* user manual. Dashed line indicates *G*mag for AT Pyx. Note that above *G*mag = 12 strehl ratio decreases dramatically.

The observations used in this project were taken on January 17th 2024, improving on a previous set of observations taken on May 15th 2017 with a lower signal-to-noise ratio on which the findings of Ginski et al. (2022) were based (see Figure 11).

The 2024 observations were taken with 7 polarimetric cycles (described in Section 2.1.2) - or 28 individual images - while the 2017 observations were taken with 3 polarimetric cycles - 12 images - with an integration time of 96 seconds/image for both sets of observations, and average seeing for the 2024 observations was 0.573, superior to the 2017 observations' average seeing of 0.676. This would appear to explain, to some degree, the better SNR of the newer observations. As AT Pyx has a *G* magnitude (ie apparent magnitude of the star seen through the broadest of the Gaia photometric bandpasses) of 13.319 ± 0.02 (Gaia Collaboration et al., 2021) - which is at the lower end of brightness that can be observed successfully with *SPHERE* (see Figure 10) - observations of AT Pyx are particularly sensitive to the quality of atmospheric seeing.

For all images, data reductions were performed with IRDAP (IRDIS Data reduction for Accurate Polarimetry, van Holstein et al. 2020) using default settings. Presented in Figure 11 are the final Q_ϕ images. After an IRDAP reduction, Q_ϕ images contain all the azimuthally polarized signal as positive pixel values and all radially polarized signal as negative values (de Boer et al., 2020). In the case of the images displayed in Figure 11 the pixel values have been corrected to represent received flux in mJy.

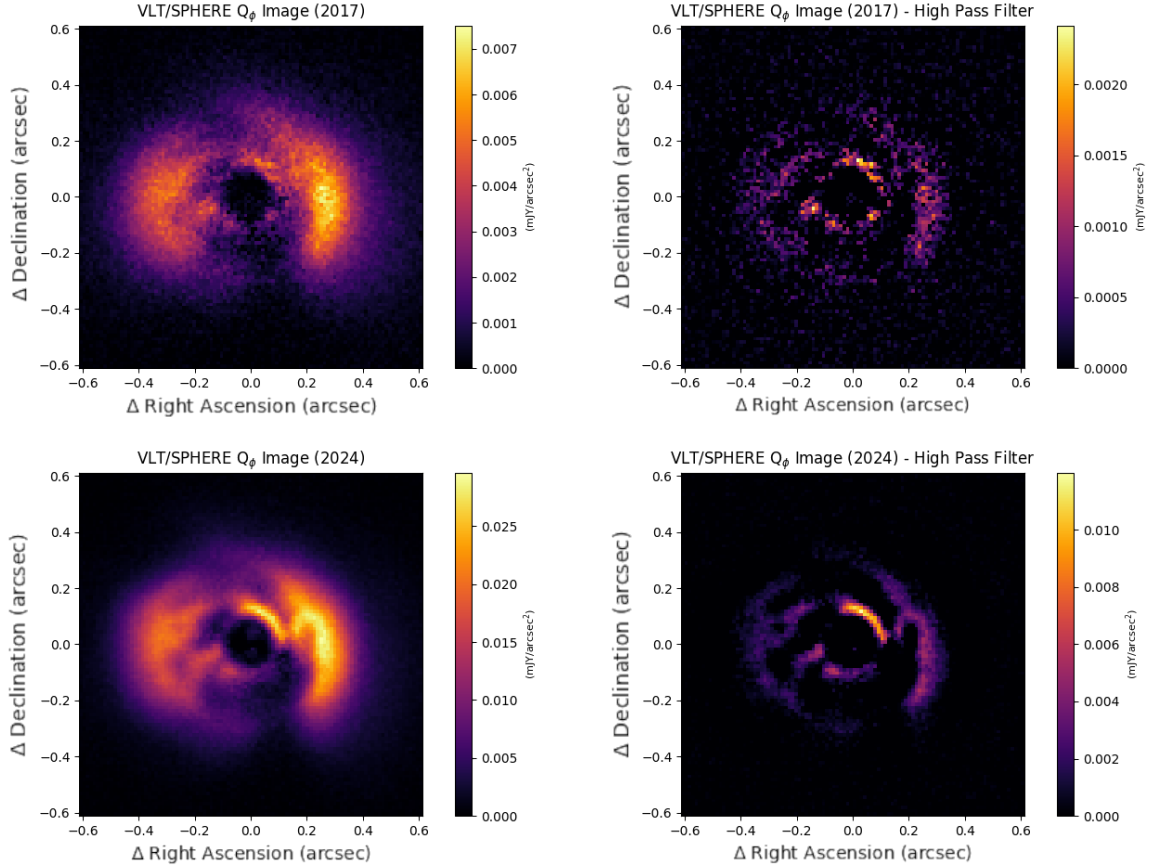


Figure 11: SPHERE Q_ϕ images of AT Pyx in the H-band. TOP: 2017 observations, BOTTOM: 2024 observations.

The SPHERE data provides a view of the ‘top’ scattering surface of the disk, with the disk inclined towards the observer about a major axis approximately 22° to the horizontal (disk orientation explored in greater detail in Section 5.1). The northern edge of the disk can be considered the near side (or front scattering side) as it is brighter, in line with the theory that the front scattering side of a disk should provide more received flux due to the scattering phase function (see Section 2.1.2). In Ginski et al. (2022) three spiral features were highlighted from the 2017 dataset, these can be identified consistently across the 2017 and 2024 datasets and are particularly clear with a high pass filter applied (see Figure 11). As all distinct disk features are identical in both observational epochs, presented in Figure 12 is the 2024 Q_ϕ image with all distinct features highlighted.

As mentioned in Section 4.2, the edges of the cavity identified in continuum emission appear to trace the arcs seen in scattered light. This could provide further evidence for AT Pyx playing host to a large empty cavity, with the arcs tracing an outer disk and the spirals launching from an inner disk.

There is also what appears to be significant shadowing in the southern (far) side of the disk, possibly as a result of the inner disk (if there is a large enough cavity for a separation between inner and outer disk) being misaligned with the outer disk, this is touched upon further in Section 7.3.2.

An extensive study into the characteristics of the observed spiral arms is undertaken in Section 6.1.1.

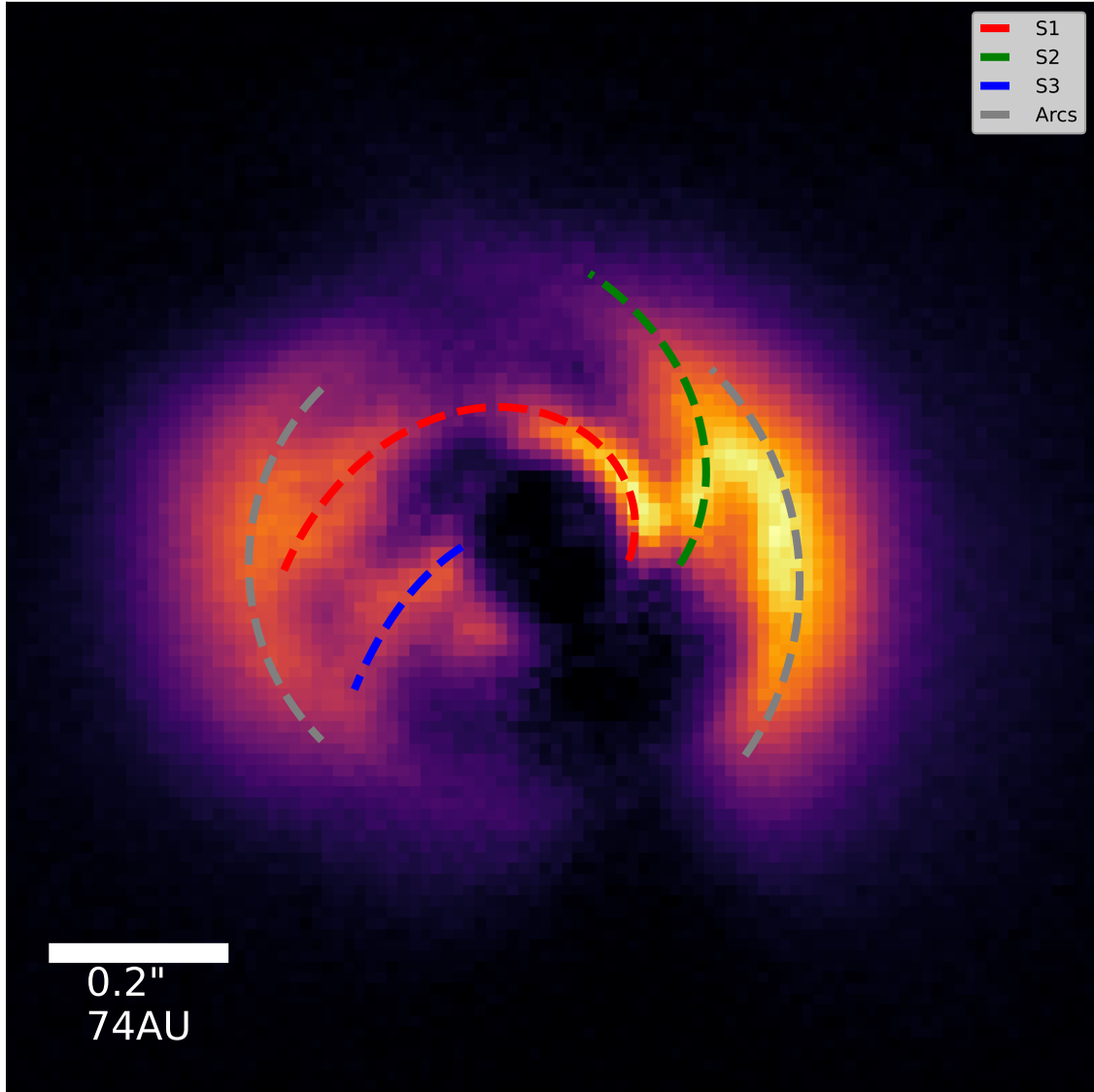


Figure 12: AT Pyx disk features. Spiral features are numbered in order of decreasing apparent length.

5 Analysis of System Geometry

5.1 Gas Velocity Structure

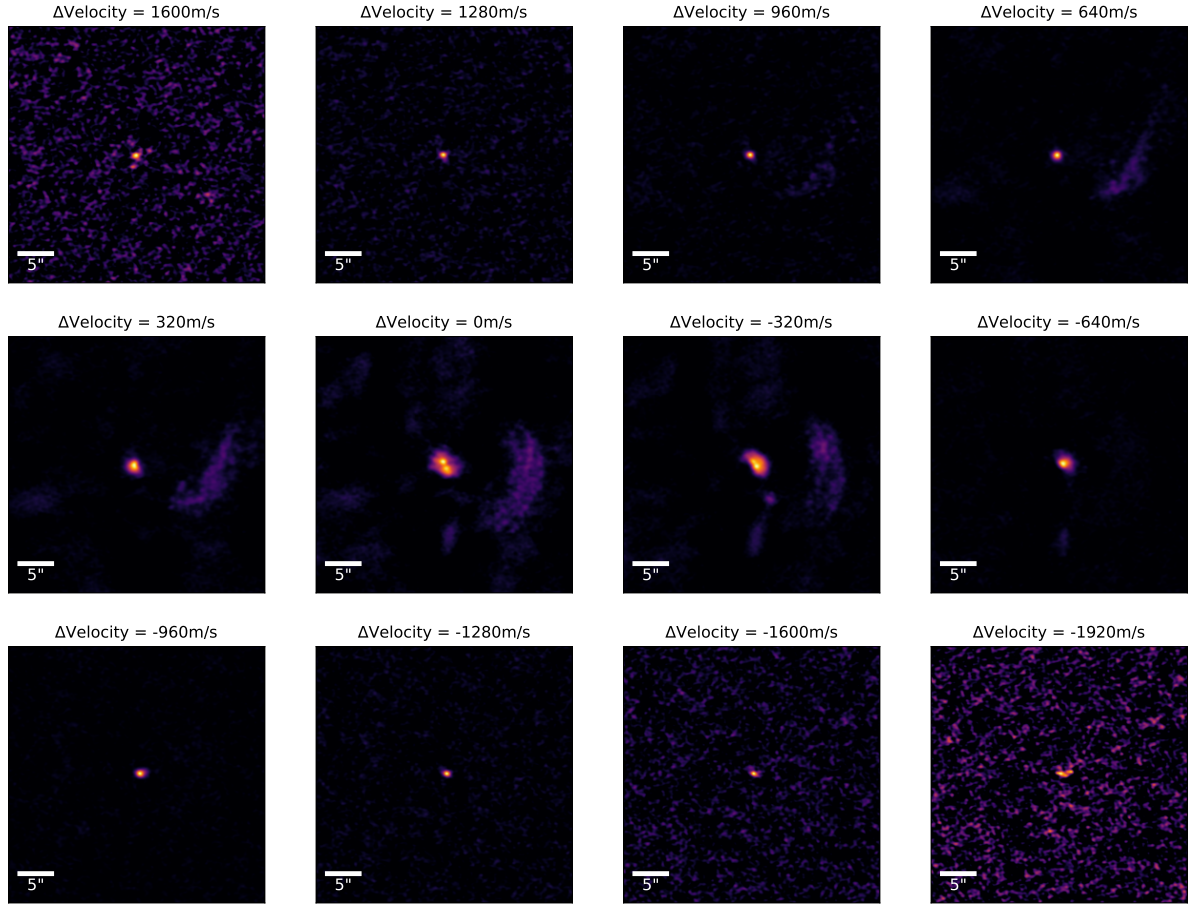


Figure 13: ALMA velocity channels in cube truncated to only channels where disk signal-to-noise high enough to distinguish disk. Note that as systemic velocity ($\Delta v = 0$) is approached, signal from surrounding cloud material is also picked up.

In order to proceed with a morphological analysis of AT Pyx’s disk - in particular for characterisation of spiral arm structure and disk eccentricity - information on the disk’s 3D structure needs to be acquired from the ALMA gas line data - most importantly values for disk position angle (angle between disk major axis and x-axis) and inclination (angle of rotation of disk plane about the major axis with inclination = 0° referring to a disk viewed top-down) which are needed for the deprojection carried out in Section 5.2.

To this end rotation maps are generated using the package Eddy (Teague, 2019) where Keplerian rotation patterns are fitted to the moment 1 maps of AT Pyx created with the BetterMoments package (Teague and Foreman-Mackey, 2018). To briefly explain moment maps, the ALMA line data is presented as a cube of images that can be interpreted as slices within a spectral range covering the velocity range of the target (a source moving away or towards an observer emits light that is red- or blue-shifted to an extent dependent on velocity, see Figure 13), in this case AT Pyx’s disk. The zeroth moment map

Table 4: Parameters and equations defining Eddy fit

Parameter	Symbol	Description
Centrepoinets	x_0, y_0	Co-ordinates of central star in rotation map in arcseconds
Stellar Mass	M_{star}	Mass of star in solar masses
Position Angle	PA	Angle of minor axis to y-axis in degrees
Systemic Velocity	v_{LSR}	Velocity of whole system in metres/second
Inclination	inc	Angle about the major axis that the disk plane is rotated in degrees
Aspect Ratio	z_0	Gas surface height h_0 at 1" for the emission surface
Flaring Power	ψ	Exponent that determines the power-law for gas surface height
Velocity		$v_0 = \sqrt{\frac{GM_* r^2}{(r^2 + z^2)^{3/2}}} \cos(\phi) \times \sin(inc) + v_{LSR}$
Emission scale height		$z(r) = z_0 \times \left(\frac{r}{1''}\right)^\psi$

displays the total intensity at each pixel over the entire range of the cube - effectively showing the gas content of the disk (within the extent to which this gas content is optically thin), where the first moment map demonstrates the average velocity in terms of redshift or blueshift at each pixel (see Figure 8).

Eddy’s rotation map algorithm operates based on fitting the relations defined in Table 4 to the moment 1 map. With ‘best guesses’ for input parameters the Eddy ‘`fit_map`’ function can be run on a moment 1 map and over several iterations the equations in Table 4 can be solved until the parameters converge, thus finding values for all parameters simultaneously.

The goal is thus to find the solution that provides the most accurate values for the *known* parameters and so the concurrent values for the *unknown* parameters must be accurate and can be used in further analysis of the higher resolution scattered light images. A position angle of $\sim 240^\circ$ can be found easily from the very first iteration of the ‘`fit_map`’ function as the minor axis sits along the boundary between redshifted and blueshifted regions of the moment 1 map due to the relative velocities at different edges of the disk as it spins, likewise a systemic velocity value of $\sim -12 \times 10^3$ m/s can be obtained immediately from the associated frequencies/velocities of the ALMA gas line datacube (see Figure 8). Accurate stellar mass values are already known from the XSHOOTER data to be within the range of 1.2-1.3 $M_\odot$.

As the two parameters that remain fixed throughout each run and are not affected by fitting are inclination and distance - and inclination is the most desired parameter to obtain - a number of runs were performed with varying input inclination. A range of inclinations between 35° and 45° (based on the values from Ginski et al. 2022) were used initially with the selected range narrowing depending on what input value produced the most accurate stellar mass. In order to investigate the sensitivity of this method, the effect inclination has on a stellar mass fit for a single pixel with fixed values for $r, \phi, \psi, z_0, \Delta v$ is plotted in Figure 14, based on the equations in Table 4. As there appears to be a reasonably steep relationship it can be assumed that this method will characterize inclination sufficiently for the scope of

this project.

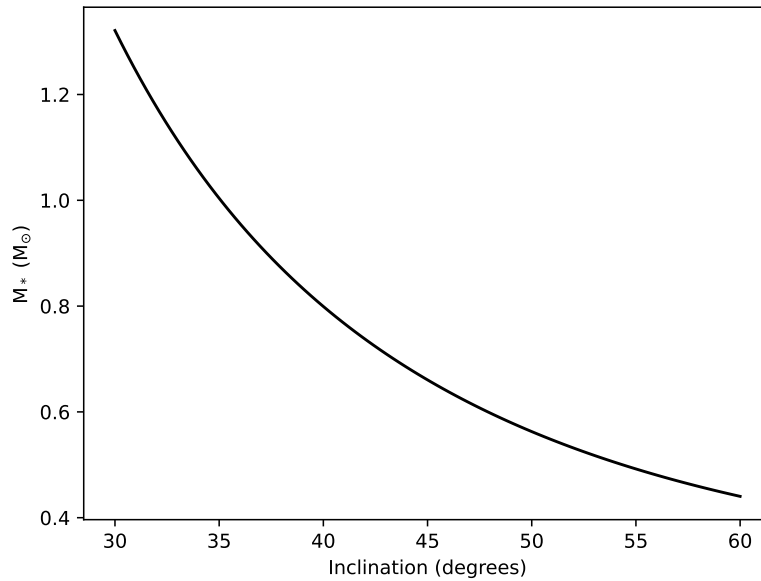


Figure 14: Relationship between inclination and stellar mass for an arbitrary pixel located at $r = 1'' = 370$ AU, $\phi = 0$ with fixed $\Delta v = 800$ m/s, $\psi = 1.2$, $z_0 = 0.4r$. Note that this plot does not refer directly to a fit to AT Pyx’s disk but demonstrates the sensitivity between these parameters for a fit to any disk.

Each run was subdivided into five iterations of the fitting algorithm in which the converged parameter values at the end of one iteration would be fed in as the inputs to the subsequent iteration so that by the end of the fifth iteration the walkers would have converged on parameter values with minimized error. Each iteration ran with 32 walkers operating over 1500 steps with a 1000 step burn-in phase.

In Figure 15 the results of the final run can be seen. An inclination of 42.5° was found to provide the closest value for stellar mass to the known range at $1.2295 \pm 0.0019 M_{\odot}$. Referring to Figure 11 this means that the disk major axis sits at $\sim 22.11^\circ$ to the x-axis and the disk itself is inclined at $\sim 42.5^\circ$ to the observer’s line of sight with the North edge being the ‘near’ side as more flux is received from this side in scattered light (as described in Section 2.1.2).

A common use of Eddy is to find agreement between deviations from Keplerian rotation patterns and observed features in scattered light. However visible tracers cannot be accurately represented due to the low spatial and spectral resolution of the ALMA data (see Section 4.2). The gas emission is only covered by a 50×50 pixel bounding box, which is only $\sim 8 \times 6$ beams. The presence of additional non-disk signal due to contamination from the surrounding globule envelope also could be responsible for the large residuals (see Figure 15); as can be seen in Figure 13, the velocity channels approaching systemic velocity play host to cloud signal, and there are not enough available channels to strip contaminated channels out without losing valuable disk signal. Future observations of this system with ALMA should aim for configurations with higher spatial and spectral resolution so as to provide a wider range of channels over which cloud contamination can be more effectively limited.

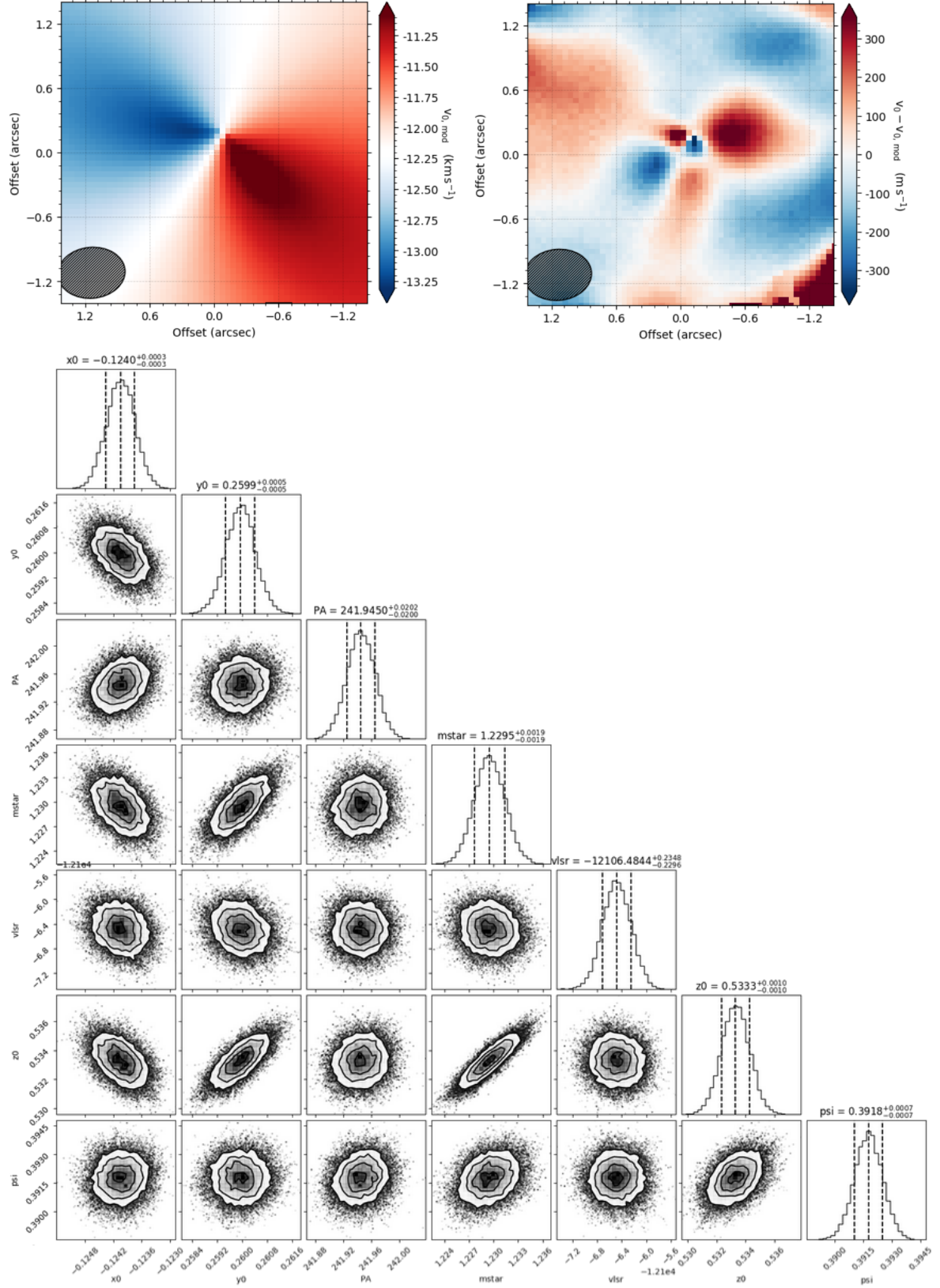


Figure 15: TOP LEFT: Rotation map showing an idealized Keplerian rotation pattern for AT Pyl's gas content. TOP RIGHT: The residuals, i.e. the rotation map subtracted from the moment 1 map to show deviations in the rotation patterns from a Keplerian velocity field. Note the high residuals (dark red) could indicate some level of pollution from ¹²CO signal in the surrounding globule gas. BOTTOM: Corner plot showing the final converged values for the fit.

5.2 Deprojection

In simulations probing properties of circumstellar disk features such as spiral and annular structures or eccentricity, particularly the ones used in Section 6, the final products are often idealised images of the disks as seen from ‘top-down’. This is a major obstacle to analysis of disks significantly inclined to the plane of the observer such as AT Pyx’s when the primary source of conclusions regarding the driving forces behind observed structures is the results of these simulations.

And as disks (in terms of small dust grains as observed in scattered light) are optically thick and not flat, following a non-linearly increasing height profile with radial distance outwards (see Figure 16), a pseudo-top-down view cannot be created merely by stretching the image along the disk major axis. Chiang and Goldreich (1997) demonstrate from hydrostatic models that a disk heated by its central star flares to adopt a vertical height profile following a power law. The disk image hence must be deprojected based on the defining parameters of this 2-dimensional power-law height profile; aspect ratio h/r and flaring angle power α .

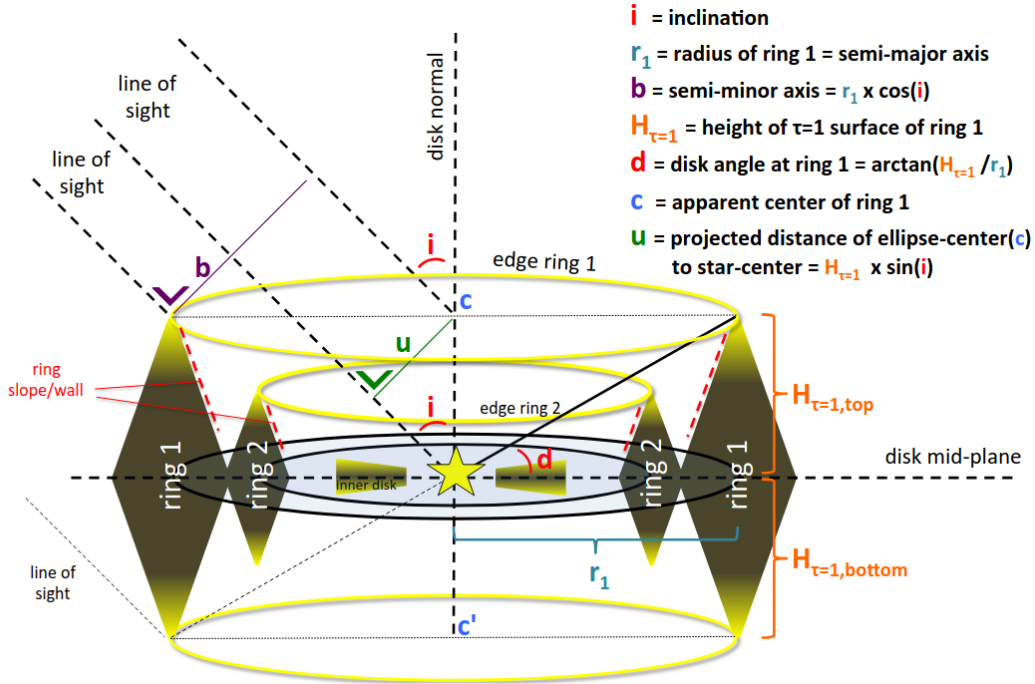


Figure 16: Figure 6 in de Boer et al. (2016) showing an approximation of disk 3-d structure with the most important parameters for deprojection being offset u and inclination i .

The vertical height profile as a function of radius can be defined as

$$h(r) = h_0 \left(\frac{R}{r_0} \right)^\alpha \quad (2)$$

with h_0, r_0 referring to the height of the disk scattering surface at a fixed distance 100 AU. Deprojection involves shifting each pixel in the image by an offset u which refers to the observed distance between a point on the disk’s top scattering surface and the point it should be observed at should the top scattering

surface be flat, sitting on the top plane seen in Figure 16. This offset can be modelled simply as

$$u = h(r) \sin(i)$$

Avenhaus et al. (2018) provide values for $(h/r)_{100AU} = 0.1617$ and $\alpha = 1.219$ from a fit applied to a survey of disks observed with SPHERE in the same conditions as AT Pyx and so for the purposes of this thesis AT Pyx’s disk is assumed to have a height profile defined by the same values. Thus, the offset by which each pixel must be moved is

$$u = 16.17 \left(\frac{R}{100AU} \right)^{1.219} \sin(i)$$

allowing a deprojection pipeline to be built for the SPHERE scattered light data.

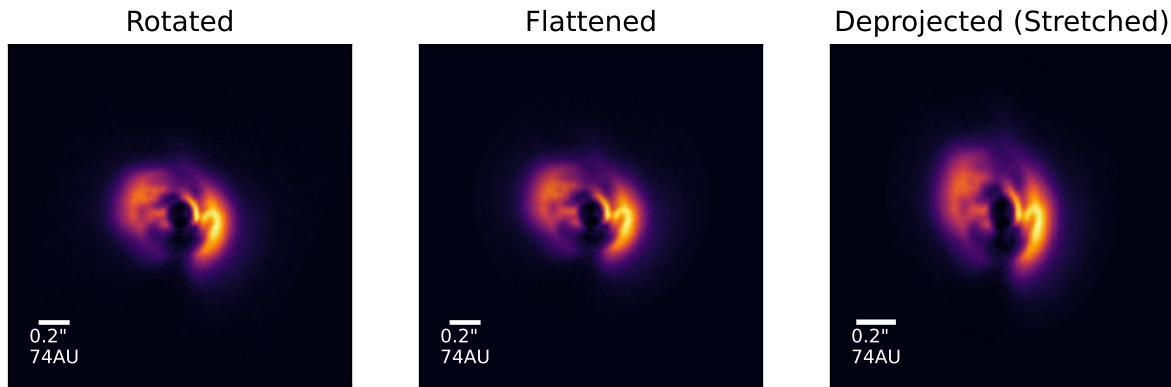


Figure 17: The deprojection pipeline in action. LEFT: The original image reoriented so that the major axis lies along the x -axis. MIDDLE: Pixel values shifted in the y -direction by offset u as a function of radial distance from the central star. RIGHT: The final image, stretched and binned down to provide an artificial top-down view of AT Pyx’s disk.

Deprojection Pipeline:

1. **Rotation:** First the image is rotated so that the major axis lies along the x -axis. This is not strictly necessary but it simplifies the whole process by limiting the amount of trigonometrical calculations involved in successfully deprojecting the disk as moving pixels by the defined offset only needs to occur parallel to the minor axis.
2. **Interpolation:** The image is stretched in both directions by a factor of ten using a spline interpolation method. This artificially increases the effective resolution of the image from 12.251 mas/pixel to 1.2251 mas/pixel or ~ 0.4533 AU.
3. **Flattening:** Each pixel in the image within a radial limit is shifted in the positive y direction (towards the front scattering side) by u (converted from AU to pixel values) effectively moving every pixel to the top plane of the disk. R is defined as distance from the centrepoint of the image *i.e.* the location of the star.
4. **Deprojection/Stretching:** The final stage now that the disk has been rendered ‘flat’ is to stretch the image along the y -axis in the traditional method of re-orienting flat images digitally. This is

done once again with spline interpolation, with the factor by which the image is stretched defined by $1/\cos(i)$. Following this the image is binned down to its original resolution of ~ 4.533 AU/pixel.

An additional challenge in the case of AT Pyx is that with the confirmed inclination and position angle from the gas line data analysis the disk satisfies the first scenario suggested by Ginski et al. (2022) which concludes that the disk would appear to be eccentric and so while usually a successful deprojection results in a perfectly circular disk, this cannot be counted on here. Thus there are no visual clues as to whether the deprojection is accurate aside from faith in the values for pitch angle, inclination angle, (h/r) and α . The output disk as seen in Figure 17 is hence eccentric, and an ellipse fitted to determine the exact eccentricity is utilised in Section 6.1.2.

Because of AT Pyx’s complexity, it is difficult to see whether or not the results of deprojection are accurate purely by visual inspection. Hence in order to validate the deprojection pipeline’s results it is tested on a disk with simpler structure. The disk around the star WISPIT 2, recently observed by van Capelleveen et al. (accepted) is a perfect test disk as it appears to have an axisymmetric circular shape with its flaring extent measurable by the offset of its multiple rings. The height profile for this disk was measured in van Capelleveen et al. (accepted) and this deprojection using the values defined by this height profile measurement was used in that publication. As can be seen in Figure 18 the disk is successfully flattened to correct for the flared surface and the final image is a near-perfect circular top-down disk image within the limitations of signal present in the input image.

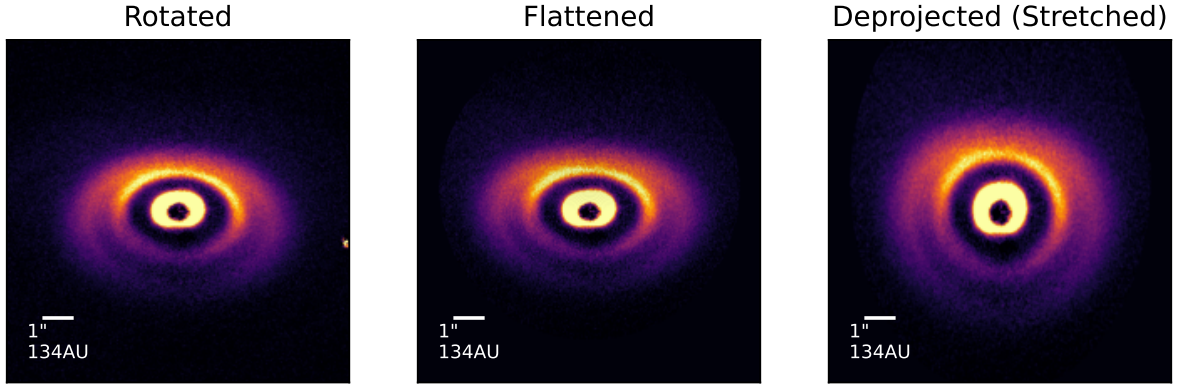


Figure 18: The deprojection pipeline tested on WISPIT 2, a more ‘well-behaved’ disk with clear flared structure. The transition from a flared disk to a flat disk to finally a circular top-down view of disk structure demonstrates that the deprojection pipeline is capable of producing accurate results.

6 Investigation of Disk Morphology

6.1 Investigation of Planet-Forming Signatures

6.1.1 Spiral Arms

The presence of spiral features may not necessarily always indicate the presence of planetary perturbbers. Large spiral arms can be generated in disks by self-gravitating forces from global gravitational instability (Lin and Shu 1964, Dong et al. 2016, Baehr and Zhu 2021) and this is investigated further in Section 7.2. In this section it is assumed that the spirals are driven by gravitational influence from a point source in order to fully investigate the planet-forming scenario. Also with new higher SNR SPHERE observations more comprehensive analysis on spiral structure can be undertaken than what was possible in Ginski et al. (2022) with the 2017 SPHERE dataset mentioned in Section 4.3.

In this section the photometric contrast ratios of the spiral arms, radially-dependent pitch angles of spiral arms and azimuthal arm-to-arm separations are used to constrain the mass of a possible embedded planetary-mass companion.

Arm Contrast

This strand of analysis draws heavily from the work of Dong and Fung (2017) in which hydrodynamical and radiative transfer simulations of disks with a range of masses of planetary perturbbers at a fixed radial distance demonstrated a relationship between contrast of spiral arms and planet perturber mass. As can be seen in Figure 19 each mass drives a signature peak contrast at either side of the planet’s orbital radius.

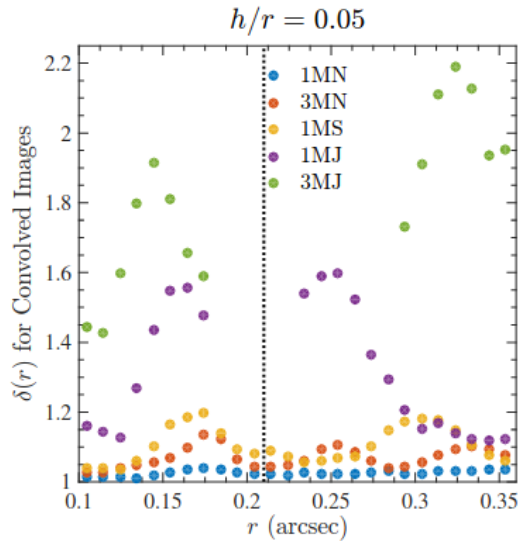


Figure 19: A set of results from Dong and Fung (2017) (figure 7) showing contrast of spiral arms as a function of radial distance, for a simulated disk with $(h/r)_{r_p} = 0.05$ and a range of masses of planetary perturber (key: MN - Neptune Mass, MS - Saturn Mass, MJ - Jupiter Mass, black dotted line represents position of planet orbit).

Contrast here is defined as

$$\delta(r) = \frac{\text{Max Surface Brightness}(r)}{\text{Azimuthally Averaged Surface Brightness}(r)}$$

and in the case of AT Pyx three features (S1, S2, S3) can be identified as spiral arm candidates (see Figure 20), so for each spiral the above equation can be modelled as

$$\delta(r) = \frac{\text{Spiral Brightness}(r)}{\text{Azimuthally Averaged Surface Brightness}(r)}$$

Hence each spiral’s contrast as a function of radial distance can be modelled and compared to provide some provisional values for planet perturber mass. A pipeline was developed for the provision of contrast values along each spiral as follows.

For each spiral arm:

1. **Fitting of Galilean Spiral:** A spiral function of the form $r = R_0 \exp(b\theta)$ is fitted to lay along the path of the observed spiral (see Figure 20). This provides pixel co-ordinates in the image to extract brightness values from⁸.
2. **Extraction of Brightness Values Along Spiral:** Using the `photutils` (Bradley et al., 2025) package’s `CircularAperture` class to simulate a 4-pixel-wide (or 0.049-arcsec-wide) aperture a flux value can be extracted at each pixel co-ordinate on the spiral. Note that these values are in real units of mJy/arcsec² as the SPHERE images have been calibrated to have accurate flux units. Each flux value has an associated radius value both in terms of co-ordinate position (R) and in terms of real radial distance (r) across the disk. The value for r is calculated by adding the offset defined in Section 5.2 to R .
3. **Extraction of Azimuthally Averaged Brightness Values:** Using `photutils`’ `EllipticalAnnulus` class a simulated elliptical annulus could return the azimuthal mean flux values at each co-ordinate radius R . An elliptical annulus was chosen so as to best circumvent issues caused by the disk’s inclination. The annulus sits with its major axis congruent to the disk major axis. Ellipse major axis length $a = R$ and minor axis length $b = R \sin(90 - inc) = R \sin(47.5)$.
4. **Plotting:** Each value for spiral brightness is divided by its associated azimuthally averaged brightness value providing an array of contrast values which can be plotted against associated r .

In Figure 20 the results of contrast measurement can be seen. An important point to note is that due to the scattering phase function (see Section 2.1.2) for a significantly inclined disk the points at which the light scattered by the top surface are most representative of the flux that would be seen from a non-inclined view are the points along the disk major axis - were the disk viewed face-on, the brightness distribution across the surface would uniformly match what is seen along the major axis. Thus the most ‘true’ values for contrast can be found at the regions around which the spirals intersect the major axis, as marked in Figure 20 by the black dashed lines.

⁸The fitted spirals in this case are ideal Galilean spirals which, as can be seen in the subsequent ‘Spiral Arm Pitch Angles’, does not strictly apply to S1. However, as in this section the purpose of fitting spirals is to provide a rough guide for aperture samples through the spiral with each aperture having a 4-pixel diameter, fitted spirals that pass roughly along the brightest spiral regions but cannot extrapolate the subsequent path of the spiral are sufficient

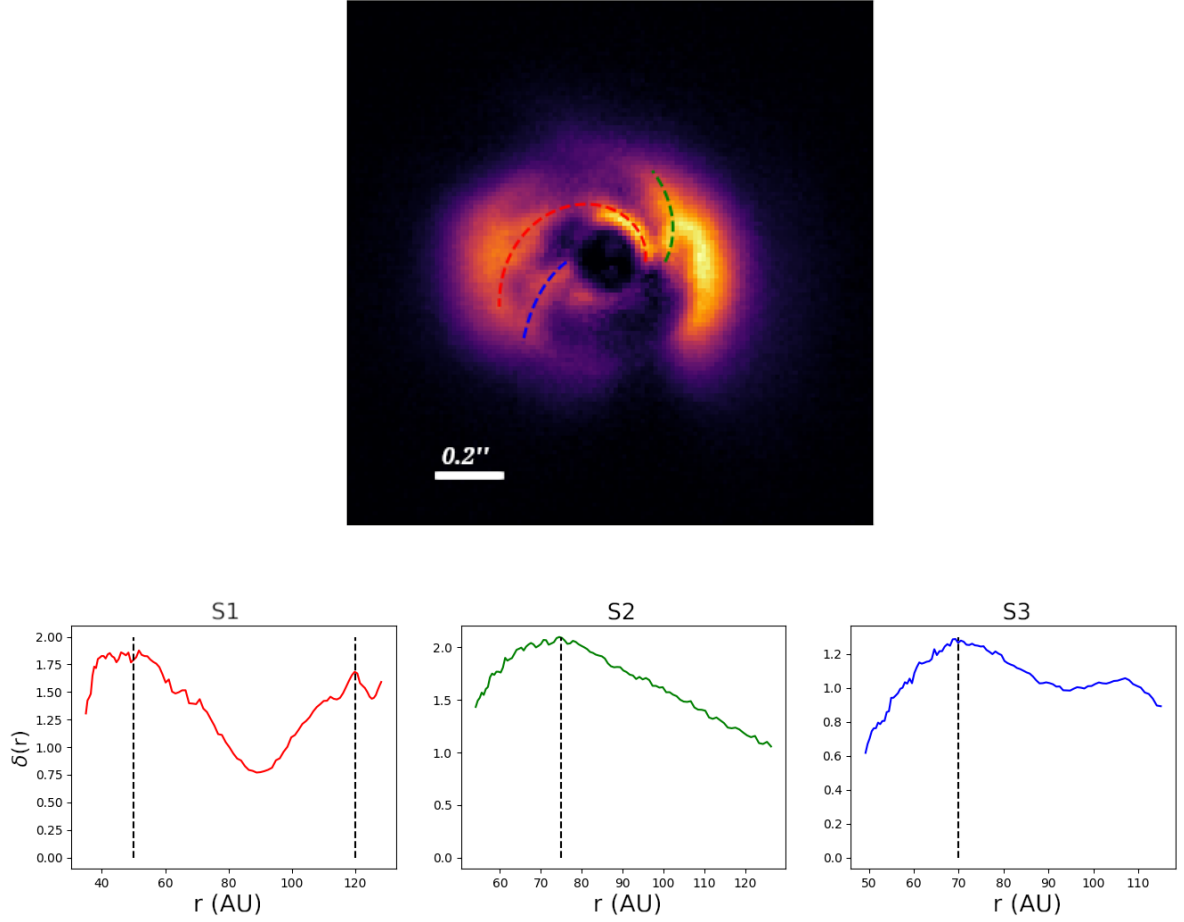


Figure 20: TOP: Spiral features in AT Pyx’s disk highlighted. Key: S1 - Red, S2 - Green, S3 - Blue. BOTTOM: Contrast for each observed spiral feature as a function of radius. Black dashed line represents points at which the spiral intersects the disk major axis.

The peak contrast values also lie in this region, meaning that they can be used for comparison with the contrast values found in Dong and Fung (2017).

S1 and S2 are in close agreement with contrast values ranging from ~ 1.6 to ~ 2 which would suggest a planet mass on the order of 2-3 Jupiter masses according to the simulation results. S3 is a slightly more complicated affair as its contrast value points towards a perturber mass on the order of 1 Saturn mass although its extreme opening angle should suggest a higher-mass perturber (Zhu et al., 2015). In fact qualitatively its extreme launching angle and pitch angle open up the possibility that it is a different kind of feature such as an accretion stream. The fact that its contrast values deviate from the values for the other two means that within the bounds of a scenario in which it is assumed all features are the result of planetary influence it could be less well-coupled to the mechanisms driving S1 and S2. Note also that S1 and S2 have very close launching points, meaning that they could possibly be a primary and secondary arm in the same spiral system, while within the observing limits imposed by the coronagraph S3 appears to launch from the opposite end of the inner disk. The nature of S3 is further discussed in Section 7.3.

Spiral Arm Structure and Distribution

In the domain of linear wave theory (Goldreich and Tremaine, 1979), planet-driven spiral arms in a circumstellar disk are understood to be formed by constructive interference between density waves with different azimuthal wavenumbers m excited by the perturbing planet at Lindblad resonances (Ogilvie and Lubow, 2002). From this model for spiral density waves, the following relation can be used to predict the shape of the gravitational wake outside planet position (R_p, ϕ_p) in polar co-ordinates (Rafikov 2002, Muto et al. 2012):

$$\phi(R) = \phi_p - \frac{\text{sgn}(R - R_p)}{(h/r)_p} \times \left[\left(\frac{R}{R_p} \right)^{1-\eta} \left(\frac{1}{1-\eta} - \frac{1}{1-\zeta+\eta} \left[\frac{R}{R_p} \right]^{-\zeta} \right) - \left(\frac{1}{1+\eta} - \frac{1}{1-\zeta+\eta} \right) \right] \quad (3)$$

with orbital frequency $\Omega(R) \propto R^{-\zeta}$ and sound speed $c_s(R) \propto R^{-\eta}$. This linear theory-based model has been shown to break down for all but very low mass ($\sim 0.01 M_{\text{Jup}}$) wave-generating planets, but on disk scattering surfaces outer spiral arms still broadly follow linear theory while inner arms are more opened than predicted (Zhu et al., 2015). In the case of AT Pyx, much more of the inner material is obscured by the coronagraph as its disk is significantly smaller on-sky than other similar disks due to its further distance, making it difficult to probe inner spiral arm structure completely but it is possible to inspect how closely the visible/outer portions of the arms follow linear theory in order to infer the location of a planetary perturber.

Furthermore with regards to spiral behaviour according to linear theory, Rafikov (2002) gives the following relationship between spiral pitch angle, scale height at planet radial distance, and sound speed:

$$\tan \theta = r \frac{\Omega(r) - \Omega_p}{c_0(r)} = \frac{r_p}{h_p} \left(\frac{r}{r_p} \right)^{\nu+1} \left[\left(\frac{r_p}{r} \right)^{3/2} - 1 \right] \quad (4)$$

From this it can be seen that spiral opening angles increase with aspect ratio and are not governed solely by mass and position of a planetary perturber.

With regards to planet-disk interaction generating spirals, M_p can be described in terms of disk thermal mass

$$M_{th} = \frac{c_s^3}{G\Omega_p} \approx 1M_J \left(\frac{h_p}{0.1} \right)^3 \left(\frac{M_*}{M_\odot} \right) \quad (5)$$

as when planet mass approaches or exceeds M_{th} such a planet can excite non-linear density waves that produce multiple spiral arms (Rafikov 2002, Juhász et al. 2015).

Azimuthal angular separation of spiral arms can also be used to infer perturbing planet properties; Fung and Dong (2015) found a scaling relationship between azimuthal primary-to-secondary arm separation as follows:

$$\phi_{p-s} = 102^\circ (q/0.001)^{0.2} \quad (6)$$

where $q = M_p/M_*$, planet/star mass ratio. This relation holds for high-mass planets located outside the observed spiral arm system and thus can be used to describe the same hypothetical planet described in the ‘Arm Contrast’ section above. This is further refined by Bae and Zhu (2018) who define a wider

range of fits in cases with varying $(h/r)_p$, and also provide limits on numbers of spiral arms generated although in the case of AT Pyx's disk low spatial resolution and obfuscation of the inner disk material by the coronagraph⁹ significantly limits any conclusions that can be drawn about the number of spiral arms in the system, this project can only speak to the 2-3 arms observed here. For an example of this methodology in practice, see Reggiani et al. (2018).

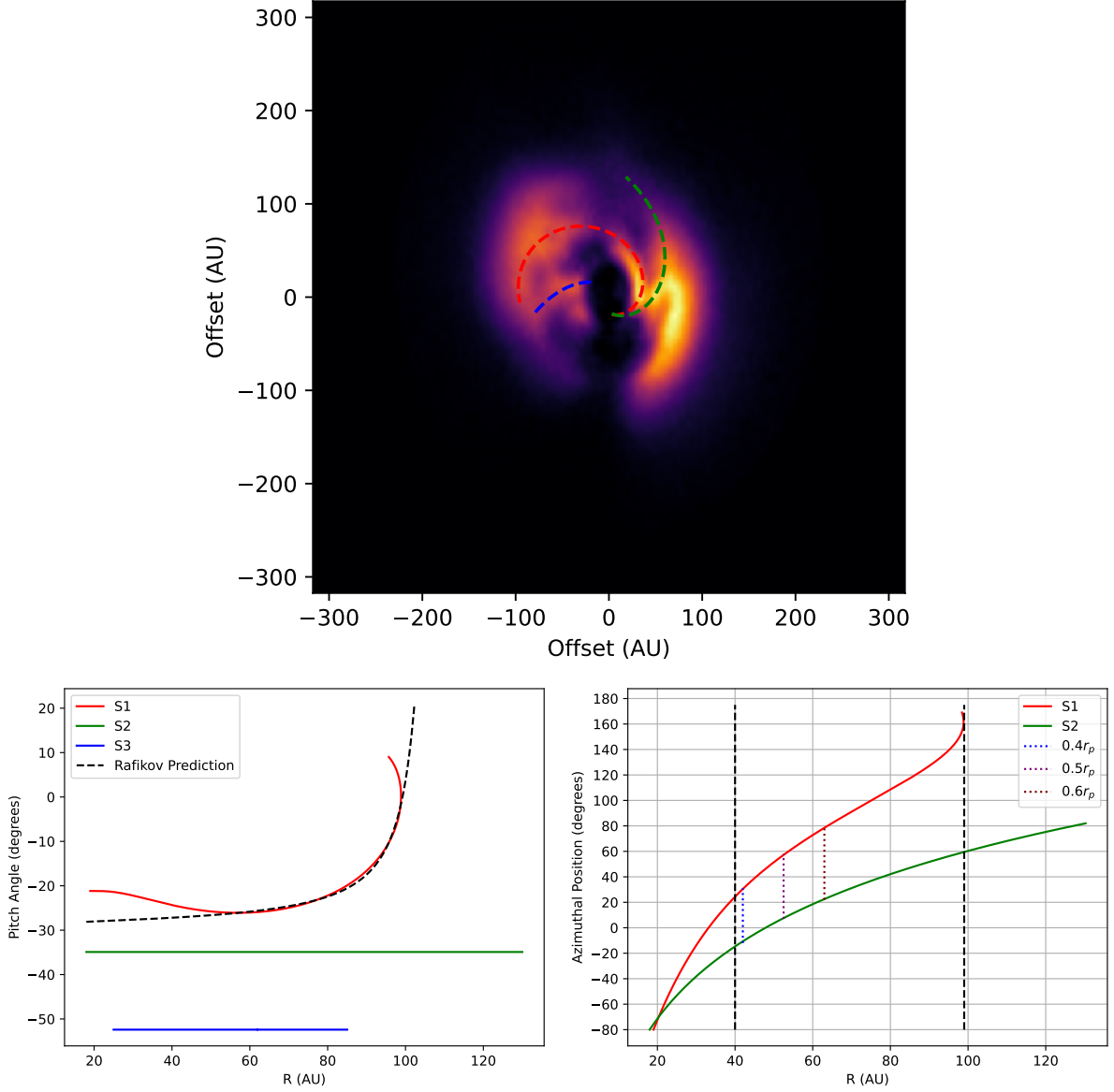


Figure 21: TOP: Spirals modelled on deprojected disk image. Key: S1 - Red, S2 - Green, S3 - Blue. S2 and S3 can be modelled as standard galilean spirals, whereas S1 deviates significantly from the default galilean spiral shape. BOTTOM LEFT: Pitch angles of the modelled spirals. Negative values tend to the left of the tangent and positive to the right. Overplotted is the Rafikov pitch angle model based on equation 4 for a planet located at 105AU with $(h/r)_p = 0.05$. BOTTOM RIGHT: Azimuthal angular separation of S1 and S2 which provides a range $40^\circ \leq \phi_{p-s} \leq 95^\circ$. Black dashed lines indicate range through which the model is tracing clear spiral signal. Dotted lines mark angular separations at positions pertaining to fits for translating separation to mass.

⁹ AT Pyx is much further away than most disks observed in scattered light - it sits just on the threshold of observability with SPHERE - and thus the coronagraph naturally obscures comparatively more of the inner disk

The deprojected spiral arms (see Figure 21) were modelled as Galilean spirals of the form $r = R_0 \exp(b\theta)$ with R_0 being distance between (0,0) - *i.e.* the star position - and observable launching point in AU. With regards to S2 and S3 the spiral formulae need no modification to fit when overplotting on the deprojected image and are as follows:

$$\text{S2: } r = 18 \exp(0.7\theta)$$

$$\text{S3: } r = 25 \exp(1.3\theta)$$

while S1 requires constraints to reshape the spiral to overplot effectively as follows:

$$r = 19 \exp(0.39\theta)$$

$$x = r \cos(\theta) + 0.2\theta^3$$

$$y = r \sin(\theta) + 0.08\theta^4$$

Additionally S3's status as a true spiral is ambiguous as it appears to follow a path almost directly radially outwards from the disk centre and arguably any curved line not necessarily following a galilean spiral path could fit it, especially when deprojected. Nonetheless its pitch angle is measured here alongside its companions.

The pitch angles of all three arms are measured as a function of radius by finding the angle between the tangent at each point and the compliment of the radial direction. This is the same property defined by Rafikov's equation (eqn. 4) and thus Rafikov's model can be fit to S1 (here assumed to be the primary arm) with the input parameters (h_p, r_p, ν) varied until the model fits. It was found that according to linear theory S1 best fits a planet position of $105 \pm 1 \text{ AU}$ with $(h/r)_p = 0.05 \pm 0.005$ and $\nu = 1/4$ as per the simulations in Rafikov (2002) (see Figure 21, bottom left).

The quality of the fit appears to be unique to these values for $(h/r)_p$ and r_p . Approximate fits with less cohesion can be made with slightly higher and lower values for both but the best possible fit applies only to the values defined above.

As S2 and S3 are modelled as logarithmic spirals, they have constant pitch angles which can be found with $\theta = 90 - \arctan 1/b$ to be 35° for S2 and 52.5° for S3.

To find azimuthal separation with increasing radial distance, the azimuthal positions of S1 (primary arm) and S2 (secondary arm) are overplotted (see Figure 21). This plot is constrained within the regions of both spirals that unambiguously trace true spiral signal within the disk. This gives a minimum separation of 40° and a maximum separation of 95° . Additionally in Bae and Zhu (2018) other fits for angular separation to planet/star mass ratio (modifications of equation 6) are provided for $r = 0.2r_p, 0.3r_p, 0.4r_p, 0.5r_p, 0.6r_p$ and in this case it is possible to retrieve separations from $0.4r_p$ - $0.6r_p$. The fits used here are

$$(h/r)_p = 0.05 : \phi_{p-s} = 117^\circ (q/0.001)^{0.29}$$

$$r = 0.4r_p : \phi_{p-s} = 109^\circ (q/0.001)^{0.22}$$

$$r = 0.5r_p : \phi_{p-s} = 104^\circ (q/0.001)^{0.19}$$

$$r = 0.6r_p : \phi_{p-s} = 104^\circ (q/0.001)^{0.14}$$

Applying the fit for aspect ratio of 0.05 to upper and lower limits of azimuthal separation 40° - 95° yields q between 2.5×10^{-5} and 4.9×10^{-4} which in turn refers to a planet mass of between 0.03 and 0.6 M_{Jup} (recall that AT Pyx's stellar mass was found to be $\sim 1.25 M_\odot$ in Section 4.1).

For $0.4r_p$, $\phi_{p-s} = 43^\circ \Rightarrow M_p = 0.02 M_{Jup}$.

For $0.5r_p$, $\phi_{p-s} = 51^\circ \Rightarrow M_p = 0.03 M_{Jup}$.

For $0.6r_p$, $\phi_{p-s} = 56^\circ \Rightarrow M_p = 0.02 M_{Jup}$.

With reference to equation 5, the thermal mass for AT Pyx's disk is $\sim 0.08 M_{Jup}$. Thus the hypothetical planet mass range found translates to between 0.25 and 7.5 M_{th} , satisfying such a planet's potential to excite multiple spiral arms.

6.1.2 Eccentricity

In Ginski et al. (2022) it was found that constraining the fit of an ellipse to encompass both the east and west ansae of the disk put the star at an offset of 55 mas from the ellipse centre, assuming a position angle of 0° between disk midplane and image x-axis (not too far removed from the 22.11° position angle found in Section 5.1). Such an offset could indicate that the disk is intrinsically eccentric.

Eccentric features in disks have been observed previously; in the cases of HL Tau (ALMA Partnership et al., 2015) and HD 163296 (Isella et al., 2016) disks were seen to have eccentric gap edges traced possibly as a result of embedded planets. Additionally Kley and Dirksen (2006) demonstrate that in 2-dimensional gas-only hydrodynamical simulations of viscous ($\alpha = 4 \times 10^{-3}$) disks with embedded high-mass planets a maximum disk eccentricity of ~ 0.25 peaking at $1.2r_p$ was found for a $5M_{Jup}$ planet after 2500 orbits.

A more in-depth study by Zhang et al. (2018) provides a range of 2-d simulations of mm dust continuum and gas surface density images which show a trend towards greater gap eccentricity with lower viscosity, lower aspect ratio and higher planet mass. The gas surface density simulations can be assumed to apply to scattered light images as small dust grains are well-coupled to the disk gas content (Takeuchi and Artymowicz, 2001).

In Section 4.2 it was shown that the ansae/arcs seen in the SPHERE scattered light image of the disk appear to trace the edges of the dip in brightness in dust continuum emission, potentially legitimizing the possibility that these ansae are the edges of a ring bordering a gap between inner and outer disks, in which case the Zhang et al. (2018) models could apply well in relating disk eccentricity to an embedded planet. However it is difficult to establish full continuity with the dust continuum image due to its low resolution.

The eccentricity of the disk was constrained by applying an ellipse fit to the outer edge of the deprojected disk prepared in Section 5.2. This was done within the framework of a program to measure disk morphologies and aspect ratios which will be presented in a forthcoming publication by Byrne et al. (see Figure 22). From this fit an eccentricity of 0.626 was found when measuring the outermost edges of the disk envelope, with the ellipse centre being offset from the star by $0.0724''$ or 26.8AU.

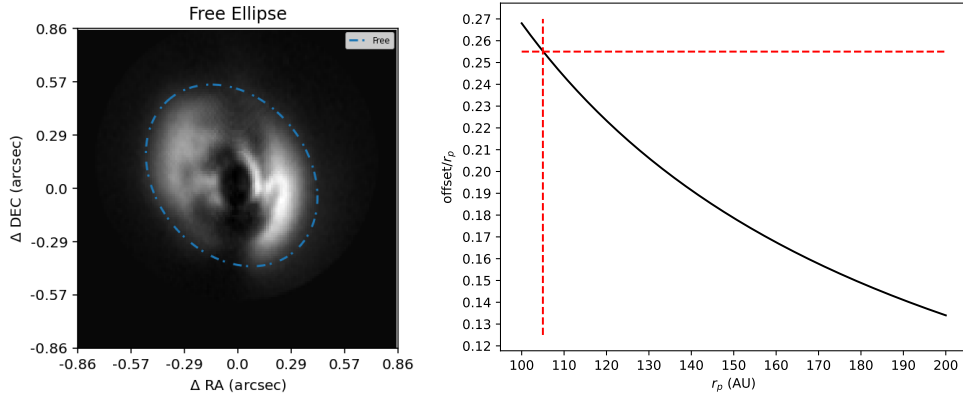


Figure 22: LEFT: Ellipse fitted to deprojected scattered light disk image. Eccentricity in this case was found to be 0.626. RIGHT: Ellipse centrepoint offset as a function of planet orbital radius r_p . Red dashed lines indicate position predicted in spiral arm pitch angle analysis. This plot derives from the results of simulations from Zhang et al. (2018) (see Figure 8 therein) relating the offset of the centrepoint of an elliptical gap opened by a high-mass planet from the central star’s position.

The eccentricity of AT Pyx’s disk is far in excess of the maximum eccentricity recorded in the simulations of Zhang et al. (2018) (~ 0.3), so this would place a very tentative mass estimate at $\sim 3 M_{Jup}$, merely because this is the maximum mass probed by the simulations. An inspection of disk ellipse centre offset from star position is slightly more conclusive (see Figure 22, right); plotting offset relative to planet orbital radius over $100 \leq r_p \leq 200\text{AU}$ gives a range 0.13-0.27. This puts planet mass prediction in the range of $3M_{Jup}$ with a disk viscosity of $\alpha \approx 10^{-3}$ and $h/r = 0.05$ (see Zhang et al. 2018, Figure 8) as these simulations are limited to discrete viscosities, planet masses and aspect ratios.

7 Discussion on AT Pyx

7.1 The Planet-Forming Scenario

As a summary of the findings of sections 6.1.1 and 6.1.2, should all distinct disk features - *i.e.* spiral arms and eccentricity - be interpreted as signatures of planet formation then the analyses in these sections yield a planet mass (assuming a single perturber) between 0.02 and 3 M_{Jup} . This value is also capped by an upper limit of 12 M_{Jup} for a small orbital radius within 0.4" and 5 M_{Jup} for an orbital radius outside the main disk complex as the K-band detection limits (see Figure 24) indicate that any planet above these masses would be detectable as a point-source in the K-band image presented in Ginski et al. (2022). Additionally fitting of a linear wave theory prediction of propagating pitch angle of the most distinct spiral arm provides a very tentative orbital radius prediction of $105 \pm 1\text{AU}$.

Of course it is not possible with this information to conclusively state that planets are present in the disk at all, and so other processes giving rise to these signatures are also investigated in the following sections.

7.2 Spiral Arms as a Result of Gravitational Instability

In terms of a non-planet origin for the spirals observed in the disk, such features can arise as a result of disk gravitational instability (Cossins et al. 2009, Dong et al. 2016, Baehr and Zhu 2021). In order to check if AT Pyx's disk is gravitationally unstable, the Toomre Q-parameter (Toomre, 1964)

$$Q = \frac{c_s \Omega}{\pi G \Sigma} \quad (7)$$

is invoked. A disk is held to be sufficiently gravitationally unstable to drive the formation of spiral arms at $Q \leq 1.7$ (Durisen et al., 2007). Following the methodology used in Ginski et al. (2025), this parameter can be found by relating gas pressure scale height to sound speed c_s : $h_g = c_s/\Omega$, where Keplerian rotational frequency $\Omega = \sqrt{GM_*/R^3}$.

The pressure scale height of the gas can be found by using equation 2, taking the aspect ratio of the near-infrared scattering surface used in Section 5.2 and dividing $h_{0,\text{IR}}$ by a factor of 4 to compensate for the different heights of the emission surfaces (Chiang et al., 2001).

Disk surface density $\Sigma = M_{\text{disk}}/2\pi R_{\text{tot}}^2$ relies on the total disk mass obtained in Section 4.2 of $M_{\text{disk}} = 9700 M_{\oplus}$. An even distribution of mass is assumed across the disk surface, which is assigned with a total extent $R_{\text{tot}} = 620 \text{ AU}$ from the ALMA ^{12}CO emission extent.

The newly-modified formula for Toomre Q-parameter is now

$$Q = \frac{2R_{\text{tot}}^2 h_0 M_*}{M_{\text{disk}} R^3} \left(\frac{R}{R_0} \right)^\alpha \quad (8)$$

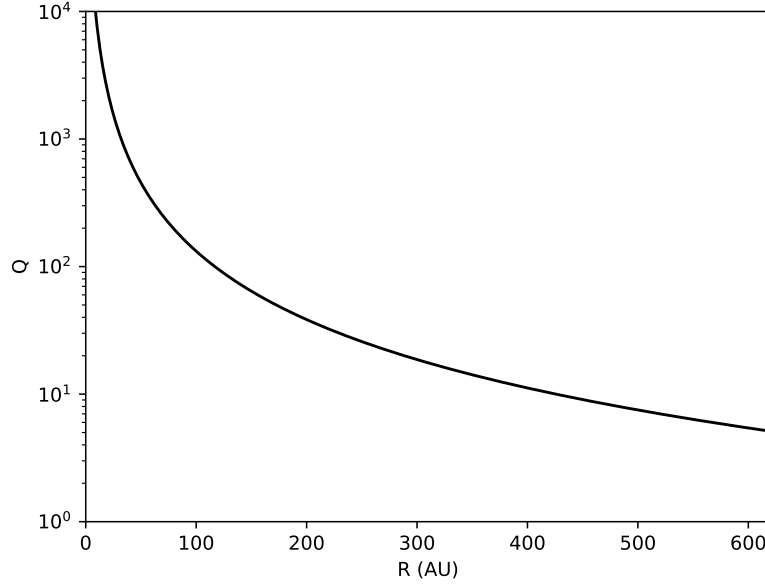


Figure 23: Toomre Q parameter for gravitational instability as a function of R .

As can be seen in Figure 23, the appropriate magnitude for gravitational instability is only approached at the furthest edges of the disk, which would seem to indicate that the spiral features of the disk are not a result of gravitational instability.

This approximation is not without flaws; an assumption of a consistent surface density across the disk overestimates the outer disk surface density, however this approximation should still indicate that the inner disk regions can be assumed gravitationally stable given the magnitude of Q for these regions would not drop significantly enough to change the result with a radially-dependent surface density model.

Another potentially significant source of error on this calculation is that the disk dust mass obtained in Section 4.2 could be underestimated, as a measurement of dust mass from integrated flux of continuum emission assumes that the disk midplane is completely optically thin. If this is not the case, then the disk dust mass could be underestimated, inflating the Q parameter.

It is found that the disk mass needs to be underestimated by a factor of 1000 for the spiral region of the disk to be within the instability regime. Hence the conclusions here about gravitational instability still hold.

7.3 The Case of S3 - Spiral, Streamer, Something Else?

The exact nature of the feature labelled S3 (figures 12, 20, 21) is difficult to determine. Its trajectory, especially in the deprojected image, appears to point radially out and intersect with the left ansa. In Section 6.1.1 it is fitted with a logarithmic spiral with a very large consistent pitch angle, which could effectively be replaced with a straight line in the deprojected image. Its lack of cohesion with spiral features S1 and S2 also limits the value of any of the techniques used in Section 6.1.1 for determining its relationship to planet formation.

In this section S3 is used as a starting point for investigations into other explanations for disk features.

7.3.1 AT Pyx as a Stellar Binary

One possibility is that S3 is a ‘streamer’, a conduit through which material from the outer disk is funnelled towards the inner disk to accrete dust mass onto the star. Such phenomena are known to occur in circumbinary disks such as GG Tau (Keppler et al., 2020), where periodic perturbations within the binary system appear to cause material to be pulled out of the outer disk towards the centre.

Hence the possibility that AT Pyx is a stellar binary is investigated, first through checking its Gaia renormalised unit weight error (RUWE) value. Given that the presence of a circumstellar disk can inflate a given RUWE parameter, the threshold for binarity sits at ~ 2.5 (Fitton et al., 2022). AT Pyx has a Gaia RUWE value of 3.532 (Gaia DR3, Gaia Collaboration et al. 2021), which is a strong indicator of binarity well in excess of the Fitton et al. (2022) limit of 2.5.

The possibility should be considered however that Gaia’s light measurements can be influenced by the dense globule in which AT Pyx is embedded; similarly to how disk material may present a level of opacity that cannot be fully probed by GAIA (Fitton et al., 2022) - thus introducing more RUWE - a sufficiently bright/dense surrounding cloud could inflate the RUWE parameter.

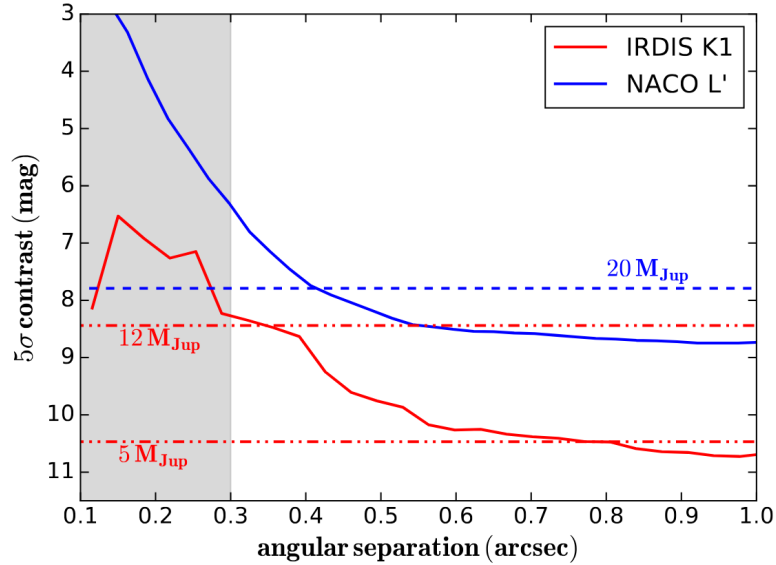


Figure 24: SPHERE/IRDIS K-band detection limit (red) as a function of angular separation from star, Figure 5, Ginski et al. (2022), grey area indicates region inside disk ansae.

A stellar companion to AT Pyx outside the coronagraph (ie outside a radial distance $r \approx 0.1'' = 37\text{AU}$) should be well in excess of the IRDIS K-band lower limit for detection as represented by Ginski et al. (2022) (see Figure 24). Additionally the XSHOOTER spectra for AT Pyx should detect lines for a second stellar object to within a separation of 0.1 AU for an object with a mass of at least $0.1 M_{\odot}$ (using the methodology from Ginski et al. 2025); XSHOOTER is capable of resolving the Doppler-shifted spectral

lines of a stellar companion down to an orbital velocity of 16 km/s, which corrected for an inclination of 42.5° gives a velocity limit of $16/\sin i \approx 25$ km/s. The Keplerian orbital velocity of a $0.075 M_\odot$ companion at 0.1 AU is ~ 25 km/s. This gives a parameter space between 0.1 and 37 AU in which an undetected stellar companion could reside.

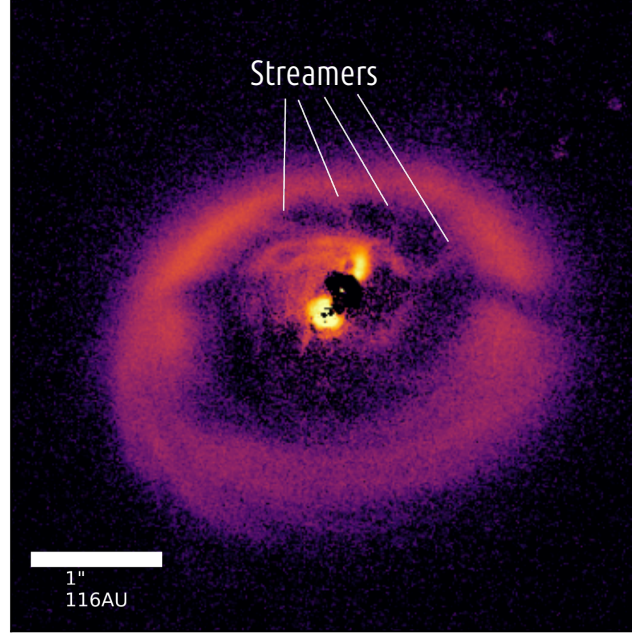


Figure 25: *SPHERE/IRDIS H-band image of GG Tau. Of particular interest is the network of streamers between the outer and inner disk as can be seen in the north side of the disk in this image.*

The unobserved parameter space along with the RUWE value mean that the possibility of AT Pyx being a binary system cannot be discounted. In which case it is compared with GG Tau (figure 25), a well-documented circumbinary disk system. GG Tau’s streamers are some of the most well-resolved examples of accretion streams between inner and outer disk and are suggested to be launched periodically by perturbations in the central binary’s orbit (Keppler et al., 2020). Within the restrictions of the resolution of the H-band images of AT Pyx, there is visual similarity between S3 and these streamers, especially how it could be interpreted as a conduit between inner and outer disk material given that it intersects the western ansa.

In order to test whether S3 moved at all between the 2017 and 2024 observations of the system, the 2017 H-band image is subtracted from its 2024 counterpart - with a Gaussian blur applied to both images in order to smooth them - to ostensibly quickly detect any major changes in the dataset (see Figure 26). In the subtracted image it would appear that the residuals are spotty and trace regions where more flux is detected in the 2024 dataset due to better adaptive optics correction, and there is no notable change in brightness in the position of S3, indicating it did not move significantly enough in 7 years for a change to be detected within a SPHERE beam (50mas) at this resolution.

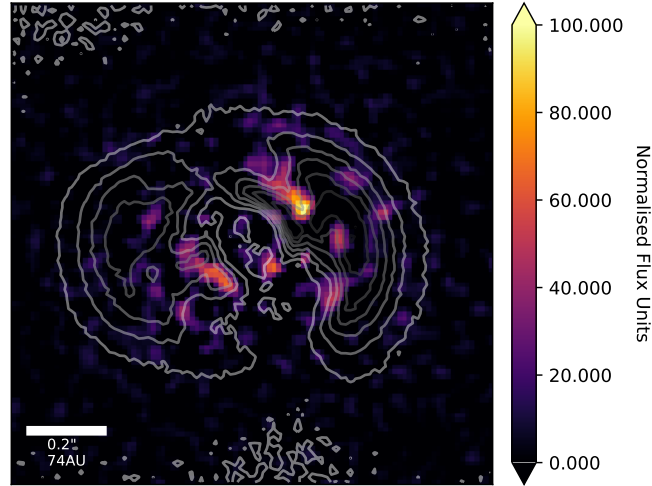


Figure 26: The result of subtracting the 2017 SPHERE H-band image of AT Pyx from the 2024 H-band image. Overlaid in greyscale is a contour plot of AT Pyx in H-band to aid with identifying positions of residuals.

If AT Pyx is a stellar binary, this could also provide an explanation for the disk’s eccentricity. Simulations by Thun et al. (2017) and Ragusa et al. (2017) find that the influence of a stellar companion, particularly with higher orbital eccentricity and mass ratio, can drive the opening of an eccentric cavity within the disk. Ragusa et al. (2017) also find that such a process can deposit overdensities of gas and dust in a horseshoe shape at the cavity boundary. The models demonstrating simulated ALMA continuum observations of such a formation (see Figure 27) appear to be qualitatively similar to the brightness asymmetry observed in the ALMA 1.3 mm dust continuum image of AT Pyx’s disk (see Figure 8).

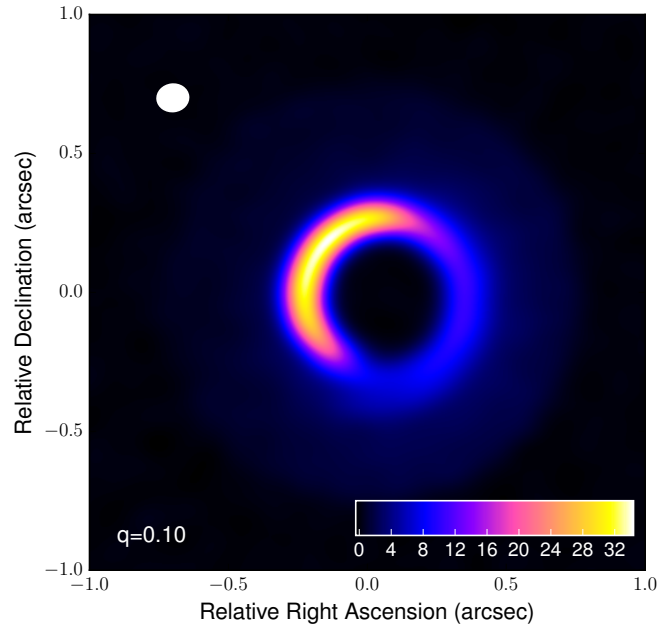


Figure 27: Figure 2 from Ragusa et al. (2017) showing a simulated ALMA dust continuum observation at 345GHz where the eccentric gap opened by a stellar binary with a mass ratio $q = 0.1$ between primary star and companion creates an overdensity of dust.

7.3.2 Cloud Capture Scenarios and Interplay Between AT Pyx and Its Kilo-AU Environment

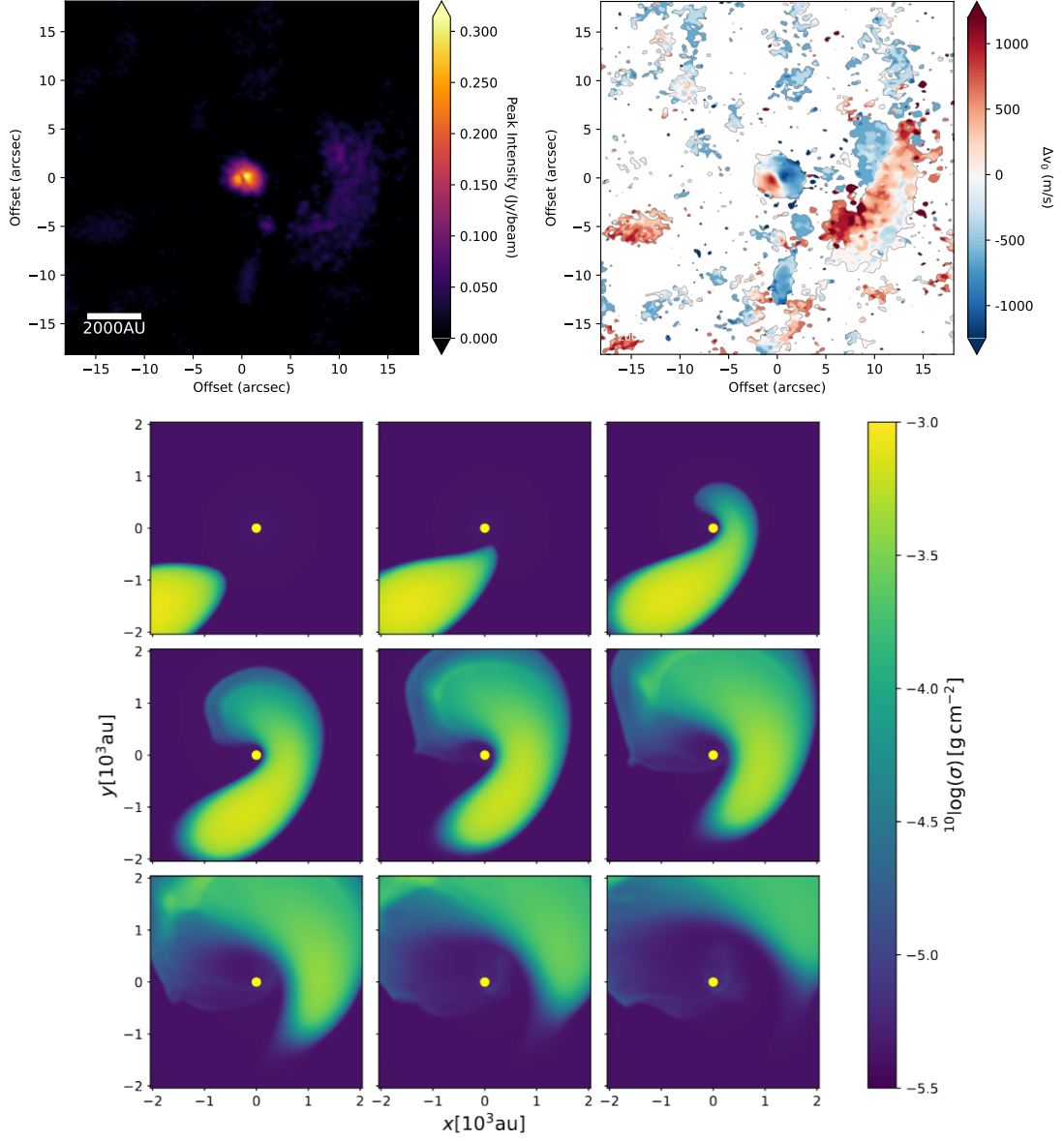


Figure 28: TOP LEFT: *Moment 8* (peak intensity) image of AT Pyx and surrounding cloudlets in ^{12}CO gas line emission from ALMA. TOP RIGHT: *Moment 1* (intensity-weighted velocity) image of the same region. BOTTOM: For comparison, Figure 1 from Dullemond et al. (2019) showing column density maps of a cloudlet undergoing capture with plots advancing from top left to bottom right in intervals of 1934 yr.

Streamers are known to be a vital part of mass accretion in the early stages of protostellar evolution (Valdivia-Mena et al. 2024, see for example Cacciapuoti et al. 2024), but these protostellar streamers do not generally survive past the transition from a protostellar object to a classical T Tauri star, dissipating with the protostellar envelope. However such features can arise visibly in the event of a late-stage infall scenario from substructural cloud material within a young star’s surrounding molecular cloud being captured by the star, and in fact their longevity can increase as the star’s velocity approaches the systemic

velocity of the background gas (Kuffmeier et al., 2019).

With regards to the effect that interplay between star and surrounding cloud can have on disk morphology, there are some qualitative similarities between AT Pyx and simulations of late-stage cloudlet capture, as well as a similar planet-disk system observed to have some continuity between the disk structure and surrounding environment. This section thus investigates both.

A tentative continuity between cloud and disk material can possibly be inferred from an inspection of the kilo-AU-scale environment around AT Pyx’s disk in ^{12}CO emission (see Figure 28); the large cloud fragment visible in the eastern side of the moment 8 image in particular appears to curl suggestively towards the disk. These formations are not unlike the simulation results presented in Dullemond et al. (2019), particularly for the simulation of a small cloud under adiabatic conditions ~ 7700 years after first stages of cloud approach (see Figure 1 in Dullemond et al. 2019, reproduced here in Figure 28), which could lend credence to the possibility that the surrounding material of CG22 has an ongoing effect on the disk structure.

To further investigate the possibility that the cloudlet seen in the kilo-AU environment of AT Pyx is infalling, the expected Keplerian freefall velocity $v_{\text{freefall}} = \sqrt{2GM/R}$ is determined. At the average cloud distance in the x-y plane - 3220 AU - v_{freefall} ranges between ~ 820 m/s (for cloud material coplanar with AT Pyx) and ~ 720 m/s (for cloud material offset in the z-direction by 3000 AU in either direction).

Unfortunately only z-velocity information can be obtained from the ALMA data, which gives a Δv_z range between -800 and 800 m/s across the cloudlet. All that can be concluded is that this is in the correct magnitude for infall velocity.

A more salient approach, based on Dullemond et al. (2019) makes use of the Hoyle-Littleton radius $R_{HL} = 2GM_*/v_\infty^2$ with v_∞ being the cloudlet velocity relative to the star. If the cloudlet size is comparable to R_{HL} then a critical impact parameter

$$b_{\text{crit}} = \frac{1}{2}R_{HL} = \frac{GM_*}{v_\infty^2} \quad (9)$$

can be defined as the approach distance at which the deflection angle of cloud matter around the star is 90° , creating a well-defined arc. In the case of the observed cloudlet, the z-velocities are used to give v_∞ ranging between 500 and 800 m/s. From equation 9, this yields a critical impact parameter within the range 1700-4400 AU (see Figure 29).

While the cloudlet at present is not spherical and these parameters apply broadly to clouds in a pre-capture state and the observed cloudlet can be presumed - if it is interacting with AT Pyx - to have already undergone capture, a circular region of cloudlet is selected and measured to have a radius of $4.2''$ or 1554 AU as a lower limit for cloudlet size as it can be assumed that applying this method to the cloud in its current state can provide a qualitative signpost for whether or not it is gravitationally bound to AT Pyx. With both the cloud fragment radius being below R_{HL} and cloudlet radial distance of ~ 3220 AU falling within the critical impact parameter range it can be assumed very likely that the cloudlet is in some stage of cloud capture.

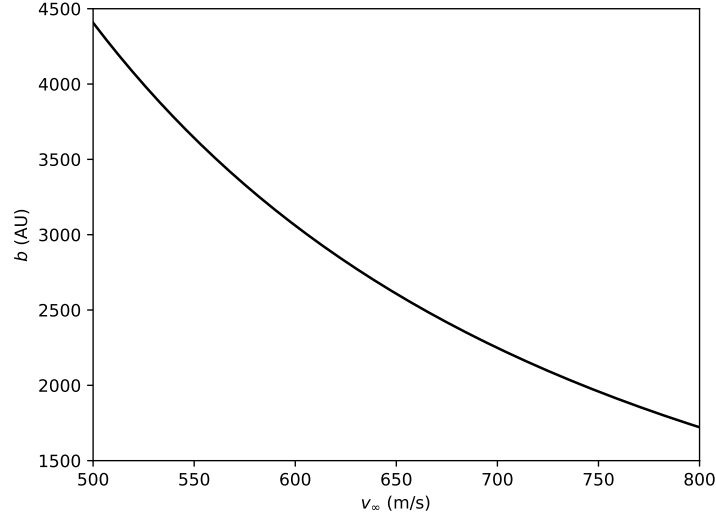


Figure 29: Critical impact parameter for AT Pyx as a function of cloud velocity relative to AT Pyx’s velocity.

In the case of the comparable system AB Aur, Speedie et al. (2025) suggest that the disk system is affected by late-stage infall - visually in the kilo-AU-scale surrounding environment it can be seen that a tendril appears to connect the disk with the surrounding cloud structure (figure 8 in Speedie et al. 2025, reproduced here in Figure 30). The infall case for AB Aur could explain the system’s formation of spiral features through the infalling material’s contribution to the angular momentum of the disk system (Speedie et al. 2025, *e.g.* Lesur et al. 2015).

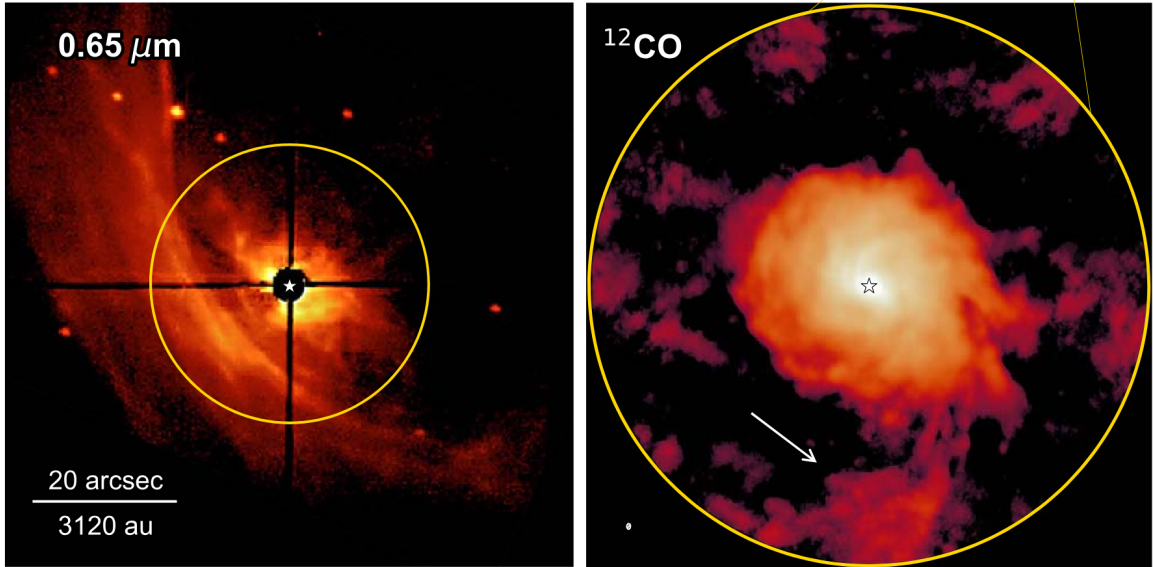


Figure 30: Figure 8 from Speedie et al. (2025) showing AB Aur’s disk appearing continuous with its kilo-AU cloud environment.

Something that could be an indicator of a significant mass contribution from infall is if the dust masses of AB Aur and AT Pyx are greatly inflated in terms of average dust masses of a known disk population. In order to check this, their dust masses as well as disk dust/star mass ratios are included among a survey

of disks in the Taurus star-forming region (Garufi et al., 2024) and a survey of disks in the Chamaeleon I star-forming region (Ginski et al., 2024) in Figure 31. Both disks are slightly, but not significantly above average in both categories among the 62 stars in the combined surveys.

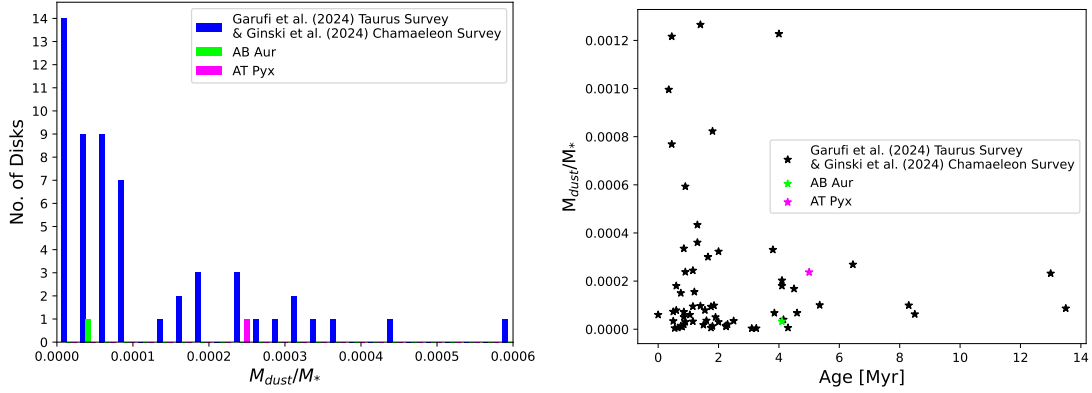


Figure 31: Disk dust mass/stellar mass ratios surveyed in Taurus (Garufi et al., 2024) and Chamaeleon I (Ginski et al., 2024) star-forming regions, with dust masses for AB Aur (Rivière-Marichalar et al., 2024) and AT Pyx (this project) inserted. Presented as LEFT: a histogram showing populations for each mass ratio regime, RIGHT: M_{dust}/M_* as a function of system age.

Presented also in Figure 32 is a visual comparison of the inner disk region of AB Aur to AT Pyx in scattered light, where the distribution and nature of spiral arms is uncannily similar, with two arms that appear to trace standard almost-logarithmic spiral morphologies and one anomalous arm with a sharp, almost-directly-radial trace. This would further support the possibility that AT Pyx’s features, particularly S3, could be a direct result of an infall scenario.

Notably this hypothesis coexists with the possibility of an embedded planet in AT Pyx’s disk as Currie et al. (2022) suggest the presence of an embedded Jovian protoplanet.

In terms of non-planet-induced eccentricity the possibility exists that AT Pyx’s disk is eccentric as a result of infall. According to simulations by Kuffmeier et al. (2021), a secondary circumstellar disk can form after the initial collapse phase due to infall from an encounter between the star and a gas cloudlet, with this secondary disk being eccentric (reaching $e \approx 0.4$). It was also shown in these simulations that for a transition disk the inner and outer disks would be misaligned as a result of this infall - misalignment of outer and inner disks can give rise to shadows observed in scattered light from the inner disk casting shadows on the outer disk.

Assuming AT Pyx’s disk is a late-stage transition disk based on the dip in brightness in the dust continuum (see Figure 9) it could be assumed that the shadowed region in the south side/far side of the disk in scattered light (see Figure 12) is a result of misalignment between inner and outer disks.

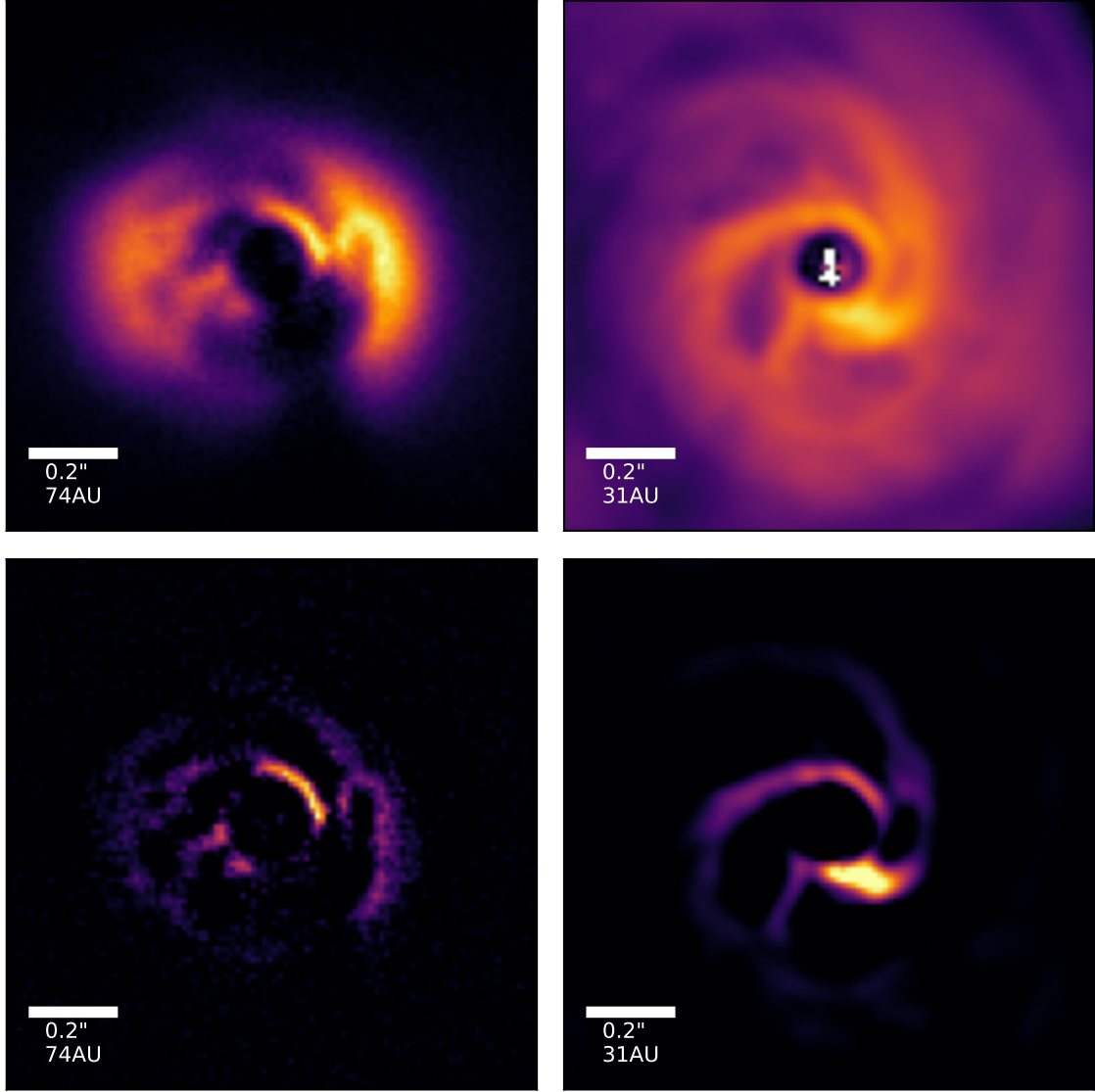


Figure 32: A comparison between the spiral morphologies of AT Pyx and AB Aur. AB Aur images have been rotated and flipped horizontally to best demonstrate structural similarities. TOP: H-band Q_ϕ images showing disk surface in scattered light. BOTTOM: Images through high pass filter to better demonstrate spiral structure. Note that AB Aur's inner spirals bear a striking similarity to AT Pyx's.

7.4 The Gum Nebula FUV Environment and Some Speculation on its Effects on Disk Formation

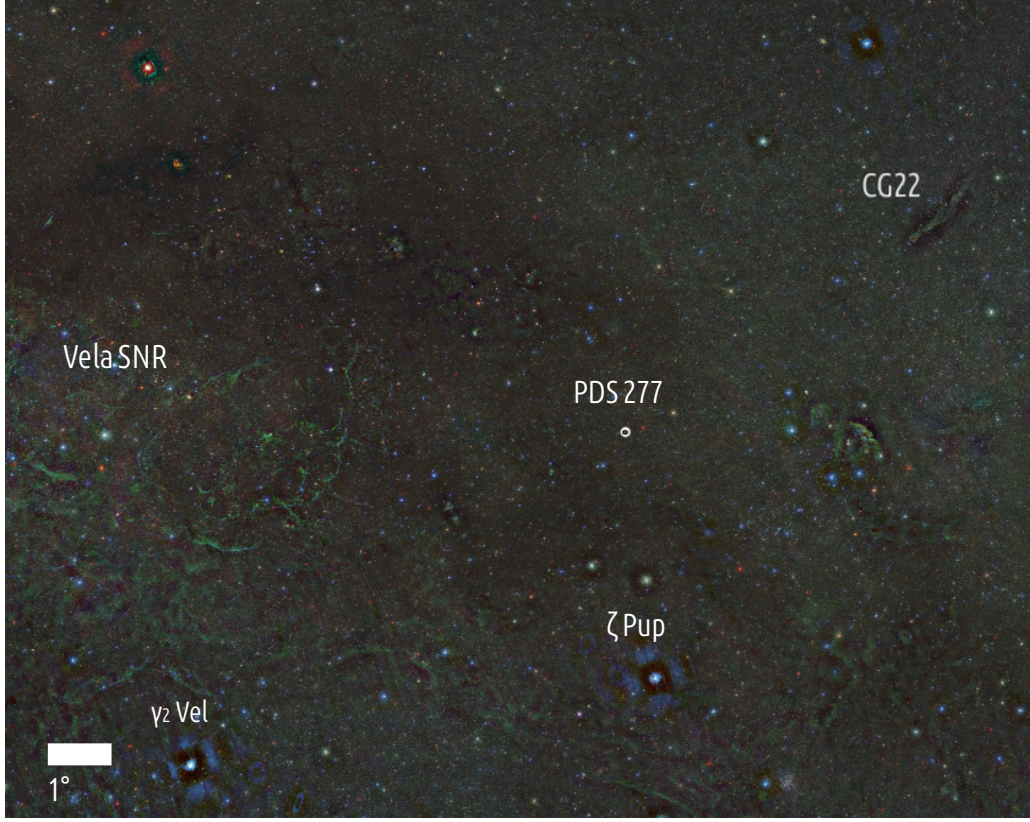


Figure 33: DECaPS DR2 (Saydjari et al., 2023) image of CG22 in its large-scale environment with FUV sources highlighted. Displayed using Aladin Lite (Baumann et al., 2022). The location of PDS 277, a star discussed in Section 7.5, is also highlighted.

On a broader scale in relation to AT Pyx’s interaction with its stellar environment, the expected influence of the ultraviolet field in the Gum Nebula is investigated in this section. While it is already suspected that CG22 must shield AT Pyx from photoevaporative FUV radiation given the lack of signatures of external photoevaporation found in XSHOOTER observations and the large disk extent, here the FUV field strength expected at such a location is calculated in order to provide an estimate of the extent to which the globule is shielding AT Pyx’s disk.

The primary source of photoevaporative FUV winds in the cometary globule complex of the Gum Nebula is held to be γ^2 Vel, a Wolf-Rayet and O star binary (Pozzo et al., 2000). The Vela Supernova Remnant’s progenitor Vela XYZ and ζ Pup, an O star, are also assumed to be contributors to the UV field that gives rise to the comet-like photoevaporative structure of the globules in the local region of the Gum Nebula including CG22, CG30, CG31, CG38 and others (Kim et al. 2005, Choudhury and Bhatt 2009). See Figure 33 for the distribution of FUV sources in the environs of CG22. Stellar characteristics of these sources are aggregated by Yep and White (2020) and displayed in Table 5.

FUV field strength here is quantified as multiples of $G_0 = 1.6 \times 10^{-3} \text{ erg s}^{-1} \text{ cm}^{-2}$, the Habing unit,

which represents average FUV flux in the ISM (Habing 1968, Winter et al. 2018).

Total FUV flux at AT Pyx’s position is obtained from

$$F_{\text{FUV,disk}} = \sum_m \frac{L_{\text{FUV},m}}{4\pi x_{\text{disk}-m}^2}, \quad (10)$$

with $L_{\text{FUV},m}$ being the FUV luminosity of each massive star and $x_{\text{disk}-m}$ being the distance between each massive star and AT Pyx.

L_{FUV} for each star was determined as follows;

1. Find bolometric luminosity from $L = L_{\text{bol,sun}} \times 10^{(4.8-M_{\text{bol}})/2.5}$ where bolometric magnitude¹⁰ $M_{\text{bol}} = M_V - BC_V$ with V-band absolute magnitude M_V obtained by converting the relevant apparent magnitude from Yep and White (2020) and bolometric correction BC_V retrieved for O stars from Pecaut and Mamajek (2013).
2. Simulate a 1-d blackbody with T_{eff} from Yep and White (2020) in the form λF_λ over the FUV wavelength range 912-2400 Å.
3. Integrate over this range using a trapezoid numerical integration method to give a proportion of FUV contribution from stellar blackbody emission.
4. Multiply bolometric luminosity by the integration result to get L_{FUV} .

In order to address the FUV contribution from the no-longer-present Vela SNR progenitor Vela XYZ, similarly to the approach of Yep and White (2020), an O star with the same characteristics as ζ Pup is placed at the location of the Vela Pulsar and put through the same calculations.

Distances between each star and AT Pyx are calculated trigonometrically with the x and y separations determined simply from on-sky co-ordinates of each source and z separations determined from differences between the Gaia (Gaia Collaboration et al., 2021) distance for AT Pyx, the distances for ζ Pup and γ^2 Vel from Maíz Apellániz et al. (2008) and the distance for the Vela Pulsar from Caraveo et al. (2001). Overwhelmingly the largest source of error on these calculations arise from the error associated with these distances, hence for each star G_0 is calculated for the smallest and largest possible separations from AT Pyx $x_{\text{disk-m}}$.

Table 5: Stellar parameters for the primary FUV sources in the region of CG22 (Yep and White, 2020) and G_0 FUV contributions from each source determined within this project.

Star Name	T_{eff} (K)	d (pc)	r ($R_\odot$)	G_0	$ x_{\text{disk-m}} $ (pc)
γ^2 Vel	57000,35000	349_{-35}^{+44}	$18.7_{-1.9}^{+2.3}, 3.2$	$0.40 \leftrightarrow 0.91$	$66 \leftrightarrow 100$
ζ Pup	39000	335_{-11}^{+12}	16	$0.43 \leftrightarrow 0.83$	$53 \leftrightarrow 74$
Vela XYZ	-	294_{-50}^{+76}	-	$0.1 \leftrightarrow 27$	$9 \leftrightarrow 152$

In current observed configurations the FUV field strength at the location of AT Pyx from equation 10 ranges between $0.93G_0$ and $28.74G_0$. When extrapolating the trajectories of γ^2 Vel and ζ Pup from

¹⁰Bolometric magnitude referring to the total magnitude corrected for reddening of the signal as received starlight passes through ISM on its path to Earth

the proper motion and radial velocity measurements provided in Yep and White (2020) (converted into units of pc/Myr in Table 6), it is also found that γ^2 Vel is currently experiencing its closest approach to AT Pyx and so its contribution to the FUV field would not have been any greater earlier in AT Pyx’s lifetime. ζ Pup has a trajectory almost parallel to AT Pyx and thus any change in distance is so small that any modelled difference in received FUV flux is within the margin of error and cannot be accurately accounted for (see Figure 34). Additionally this extrapolation neglects any changes in luminosity due to stellar evolution.

Table 6: Velocity vectors for FUV source stars in pc/Myr. All velocities relative to Earth.

Star Name	v_x (pc/Myr)	v_y (pc/Myr)	v_z (pc/Myr)
γ^2 Vel	-48.24 ± 0.13	27.09 ± 0.15	-24.4 ± 2.97
ζ Pup	-10.27 ± 0.5	17.65 ± 0.53	-15.3 ± 3.17

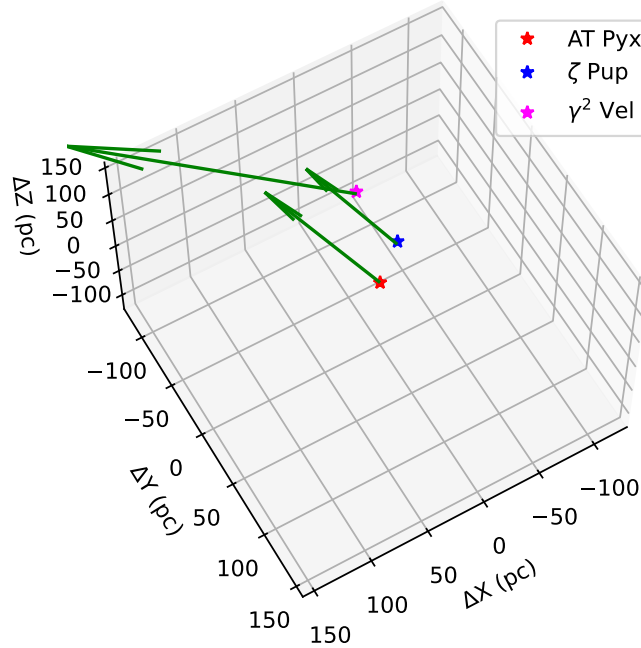


Figure 34: Paths taken by AT Pyx, ζ Pup and γ^2 Vel through space extrapolated back 5 Myr, i.e. just pre-birth of AT Pyx.

These results would in fact place AT Pyx at the lower edge of disks grouped by G_0 , when inserting it among a survey of disk-hosting stars in the Orion star-forming region by Vælgård et al. (2024) (see Figure 35). This survey observes no disks at FUV fields higher than $300G_0$, but aside from this there appears to be no trend between higher FUV field strengths and disk extents.

Under the assumption that cometary globule structure is a result of radiation driven implosion under a strong FUV field as described in Section 3.1, the dense head and distinct tail structure of CG22 - AT

Pyx's host globule - would suggest that the FUV field strength at its location is sufficient to induce some level of photoevaporation, but at this point it is difficult to determine what role FUV radiation plays in disk structure for AT Pyx, or even to what extent its position inside the head of CG22 affects its received levels of FUV flux.

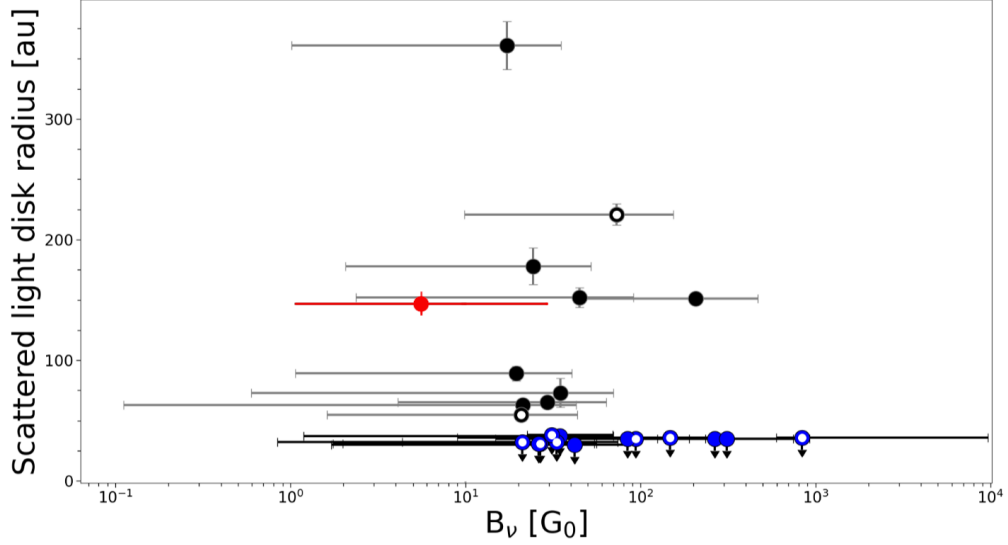


Figure 35: Figure 6 in Vægård et al. (2024) showing a sample of circumstellar disks in the Orion star-forming region with AT Pyx (red) superimposed. No trend observed.

7.5 Planet Formation in Similar Environments - PDS 277

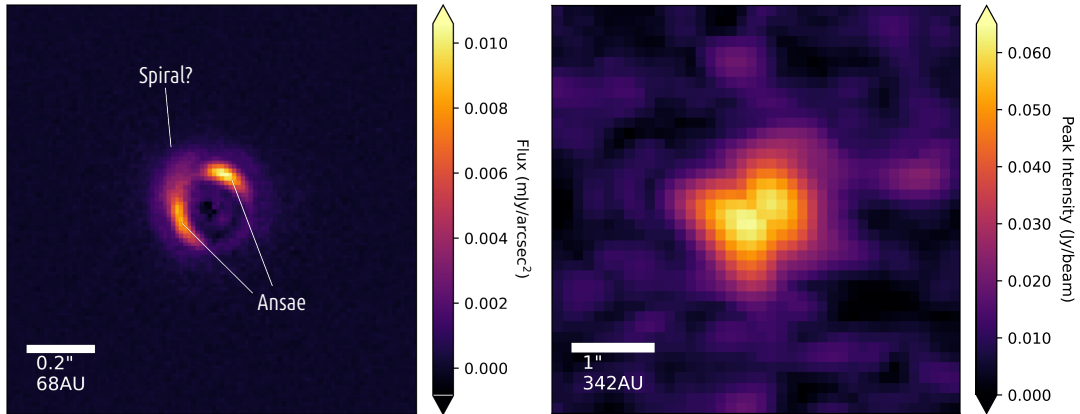


Figure 36: LEFT: SPHERE Q_ϕ H-band image of PDS277's disk in scattered light. RIGHT: ALMA moment 8 (peak intensity) image of PDS277's disk in ^{12}CO emission.

A further study into the potential for planet formation in the Gum Nebula and its globules will require a larger sample size of observed disks for comparison of all the features and properties discussed in this project. At the time of writing, SPHERE scattered light H-band data and ALMA ^{12}CO gas line and 1.3 mm continuum emission data exist for one comparable target, the disk around a similarly-located F-type

Herbig candidate star (Vieira et al., 2003) PDS 277. This system is located at a similar distance (~ 342 pc, Gaia Collaboration et al. 2021), far outside any of the globules in the region, between CG22 and the Vela OB association on-sky (see Figure 33). The SPHERE Q_ϕ scattered light image seen in Figure 36 was prepared with IRDAP (van Holstein et al., 2020) using default settings and the CO peak intensity ALMA image (same Figure) was prepared using the BetterMoments package (Teague and Foreman-Mackey, 2018), both methods having already been used extensively on the data for AT Pyx as detailed elsewhere in this thesis. Unfortunately the dust continuum image is not sufficiently resolved to determine if there is any brightness distribution comparable to AT Pyx.

Using the same methodology as in Section 7.4, the FUV contribution from the Vela OB association (γ^2 Vel and ζ Pup) ranges from $2.5G_0$ to $4.2G_0$, between 2 and 3 times stronger than its contribution to the FUV field at AT Pyx. While this is *not* anywhere near a large enough sample size of disks in this region to draw any salient conclusions, it is interesting to note - along with the increased exposure to FUV radiation from being closer to the emitting sources and unshielded by any globule envelope - that its radial extent is approximately half that of AT Pyx’s disk in scattered light, and in gas line emission approximately three times less extended.

In terms of planet-forming signatures, there appears to be a spiral feature emerging from the north-west region of the disk in the image plane, highlighted in Figure 36. This is assumed to be the front scattering/near side of the disk, given the position of the bright ansae which are assumed to constrain the orientation of the disk major axis, and the higher flux from this region in comparison to the opposite end of the disk (referring to the characteristic brightness distributions of polarimetric disk images described in Section 2.1.2). This inferred disk orientation would appear to agree with the apparent disk orientation seen in the gas line peak intensity image (figure 36, right).

This spiral feature appears to have the most in common with spiral feature S2 in AT Pyx’s disk, with its path launching from the bounding ring defined by the ansae and curling into the forward-scattering side. Fitting a logarithmic Galilean spiral to this feature yields a consistent pitch angle of $\sim 26^\circ$, which falls slightly short of S2’s pitch angle of 35° , indicating that if S2 is assumed to be the secondary arm of one spiral system perhaps the feature observed in PDS 277 could be the most extended portion of an unobserved spiral arm that begins in the inner disk. Additionally the high contrast of the ansae could suggest that they are a signpost for the border between a large vacant gap and the outer disk ring. Thus it is suspected that a similar morphological study to this project’s treatment of AT Pyx could provide evidence of planet formation within.

8 Summary and Conclusions

8.1 Conclusions on AT Pyx

This project represents a multifaceted and thorough case study of AT Pyx’s circumstellar disk using data from ESO facilities ALMA, SPHERE, XSHOOTER and ESPRESSO probing its small dust grain, large dust grain and gas content and central star stellar parameters. This study finds - through Keplerian rotation map fitting, photometry, morphological study, image processing and comparison to theoretical models - the following conclusions:

- *Disk on-sky orientation:* The disk is found to have an on-sky position angle of $\sim 22^\circ$ with front scattering/near side north of the major axis inclined at an angle of $\sim 42.5^\circ$. This is determined in Section 5.1.
- *AT Pyx as a late-stage transition disk:* From comparison between ALMA 1.3 mm dust continuum emission and SPHERE scattered light imaging in Section 4.2 it is found that there is a drop in brightness indicative of a cavity in continuum emission that the ansae in scattered light appear to trace. This would be an argument in favour of AT Pyx being a late-stage transition disk, with an inner and outer disk joined by the observed spirals.
- *Possible origins of the disk’s most distinct features i.e. spirals, rings, eccentricity:* The spirals do not appear to be a product of gravitational instability as determined in Section 7.2. Spiral features S1 and S2 - from analyses of contrast, pitch angle and azimuthal separation in Section 6 - fit the criteria to possibly be a result of planetary influence. The feature S3 is interpreted in Section 7.3 to possibly be a ‘streamer’; either an accretion stream between inner and outer disks created as a result of perturbations within a central binary’s orbit, or a streamer resulting from a late infall scenario.
- *Effect of the Gum Nebula’s moderate FUV environment on disk characteristics:* AT Pyx’s disk’s response to its FUV environment is unclear. As mentioned in Section 4.1, there are no neon lines detected in the ESPRESSO data, which indicates that very little or no external photoevaporation is taking place within the disk. In Section 7.4 it is determined that the FUV field strength at the location of AT Pyx ranges from 0.93 to 27 G_0 , which is on the lower end of FUV field strengths compared to a survey in Orion by Væglård et al. (2024). When extrapolating the paths of the two primary O stars in the Vela OB association (the drivers of FUV winds in the Gum Nebula) back to the birth of AT Pyx it is found that their contributions would have been less than today. Vela XYZ - the progenitor of the Vela supernova remnant - plays a role in the past FUV environment that is out of the scope of this project and its past contributions to the FUV field at AT Pyx cannot be concluded on here. Hence the FUV field’s effect on AT Pyx remain inconclusive.
- *Binarity:* AT Pyx has a high Gaia RUWE - used as an indicator of binarity - of 3.532, in excess of the limit for disk-hosting systems of 2.5 from Fitton et al. (2022). Although it is conceivable that AT Pyx’s presence within CG22 has a similar effect to disk material of inflating the RUWE parameter, this can be considered a strong indication of binarity. In Section 7.3.1 it is found that

there is a parameter space between 0.1 and 37 AU within the disk in which a second stellar object could reside. An assumption of binarity lends credence to the nature of S3 as an accretion stream between inner and outer disk, much like the streamers observed in GG Tau.

- *Late-stage infall:* From qualitative and quantitative comparison between AT Pyx’s kilo-AU environment and simulations in the literature of late-stage cloudlet capture in Section 7.3.2 there is evidence that AT Pyx is currently undergoing some form of late stage infall, which could be interpreted as a driving force for the disk’s spiral features and eccentricity. Additionally comparison with the system AB Aur, which is understood as a prominent candidate for late infall, yields striking similarities not only in its kilo-AU environment but also in its inner spiral features, which interestingly coexists with the hypothetical presence of a planet as AB Aur is heavily suspected to host a planet itself (Currie et al., 2022).
- *If a planet has already formed within the disk, planetary characteristics necessary to drive the observed features:* An extensive study into eccentricity, spiral morphology and brightness of observed spiral peaks in Section 6.1 when comparing measurements to simulations constrains the possible mass of a single planetary perturber needed to drive the observed features within a range 0.02-3 Jupiter masses, with a tentative orbital radius for such a perturber found from fitting a linear wave theory model to the pitch angle of S1 - the most extended spiral - at ~ 105 AU. With current observing capabilities (see Figure 24) a planet matching these characteristics cannot be detected.

AT Pyx has the potential to be a singly foundational system in the field of planet formation as it represents a consolidation of several uncommon aspects among the catalog of observed disk-hosting systems. Most studies of disks in scattered light are undertaken in surveys of low-mass star-forming regions (for instance Garufi et al. 2024 and Ginski et al. 2024), whereas along with some sources in Orion (Valegård et al., 2024) AT Pyx represents a first step into studying planet formation in other types of environment. AT Pyx’s apparent isolation from other young stars from its formation environment being a comparatively small cometary globule rather than a larger-scale star-forming region also sets it apart from stars observed as part of these surveys.

AT Pyx’s high disk mass (see Figure 31), measured for the first time in this project, is one feature that could be a direct consequence of its formation environment. That CG22 does not hold any other stars could point towards the mass budget of the cloud in the early stages of protostellar core accretion being reserved exclusively for AT Pyx, and the conclusions within this thesis about the likelihood of ongoing infall taking place may indicate that AT Pyx’s somewhat inflated disk mass is inextricably tied to its formation environment.

Also the conclusion in this thesis that AT Pyx can be modelled accurately as an eccentric disk sets it apart from most disks surveyed in scattered light. The eccentricity hinted at in scattered light is further underlined by the presence of distinct brightness asymmetries in both ALMA gas line and dust continuum emission images. This eccentricity could be interpreted to be a result of binarity, infall or the presence of a perturbing planet, with all three scenarios providing compelling evidence in their favour.

8.2 Future Work

It would appear that this project has addressed much of what can be determined within the limitations of existing data on AT Pyx. Whether it is truly a planet-hosting system is beyond the scope of what can be confirmed here, and further study on AT Pyx would require higher resolution data. With higher spatial and spectral resolution ALMA gas line data more information could be obtained regarding how structures observed in gas line emission relate to those observed in scattered light. Additionally higher spectral resolution would allow for a more in-depth analysis of the velocity profiles of surrounding cloud material for probing the late infall scenario.

With regards to higher resolution scattered light images, this project argues that AT Pyx is a prime target for future observations with the ELT (ESO Extremely Large Telescope). The enhanced resolution and sensitivity of this incoming facility will be able to shed light on many of the remaining open questions about AT Pyx, and possibly confirm whether or not there exist embedded planets within the disk.

Table 7: Targets for proposed survey of Gum Nebula YSOs. Includes samples inside and outside of globule/cloud material. All samples have identified mid-infrared excess in SEDs. All UCAC4 stars in proximity to each other. Object types truncated to Simbad (Wenger et al., 2000) otype identifier.

Star Name	RA	Dec	Object Type	Comment
CD-47 3307	07:47:04.981	-47:30:08.043	TT*	Near Vela OB Association
CD-44 3318	07:19:28.288	-44:35:11.235	Ae*	Embedded in ISM cloud
TYC 7669-1374-1	08:22:06.552	-42:34:52.585	YSO	Embedded in surrounding cloud
CD-42 4714	08:51:06.552	-43:26:11.729	YSO	Near Vela SNR
CD-42 4255	08:26:55.672	-42:49:57.221	YSO	
UCAC4 236-016587	07:36:37.516	-42:49:47.833	YSO?	Near Vela OB Association
UCAC4 239-017342	07:42:34.058	-42:13:53.361	YSO?	Within CG material
UCAC4 240-017733	07:42:16.578	-42:06:11.354	YSO?	In tail of CG10
UCAC4 243-017333	07:35:13.900	-41:25:20.189	YSO?	
UCAC4 244-017264	07:39:47.919	-41:13:41.871	YSO?	
WRAY 15-185	08:15:25.931	-34:58:24.581	Em*	Near CG30/31/38 complex
TYC 7133-1793-1	08:12:36.808	-35:38:36.149	YSO	Near CG30/31/38 complex
TYC 7130-762-1	08:16:15.311	-34:49:37.072	YSO	Near CG30/31/38 complex
WRAY 15-182	08:13:56.817	-44:34:23.237	Em*	Near Vela SNR
GSC 08138-01250	07:47:22.459	-47:00:20.205	Em*	Near Vela OB association

In terms of a broader scope of analysis, a SPHERE and ALMA survey of disks in the Gum Nebula would prove beneficial to properly address the question of how existing in this moderate FUV environment affects disk development. Particular focus on contrasting disks inside and outside other globules would also be beneficial to this end.

An ESO proposal is being prepared to carry out a SPHERE scattered light survey of other disks

located in the Gum nebula region with a focus on contrasting the appearances of disk-hosting young stars inside and outside cometary globules and molecular clouds, all with some degree of proximity to ζ Pup, γ^2 Vel and the Vela Pulsar. This survey had been submitted to ESO but was ultimately not granted time. With the feedback from this submission attempt, another proposal is being refined and prepared for submission. Table 7 includes all target stars for this proposal, and Figure 37 shows all target on-sky positions.

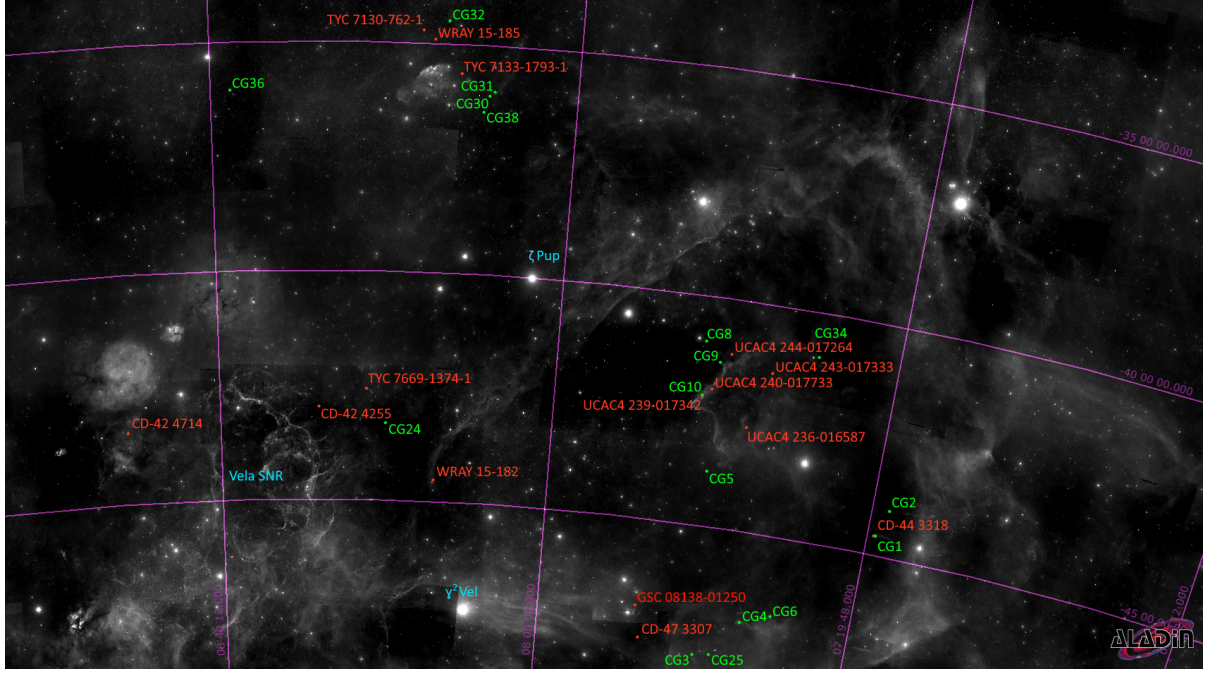


Figure 37: DSS-2-red (Digitized Sky Survey 2, Lasker et al. 1996) sky image displayed using Aladin Lite (Bau-mann et al., 2022) with all survey targets (red), cometary globules (green) and FUV sources (blue) highlighted.

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